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Review

Advances toward self-healing coatings on Mg alloys for active corrosion protection ☆

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Abstract

Magnesium (Mg) alloys with high specific strength, light weight, and natural biodegradability are promising candidates for applications in automotive industry and biodegradable medical devices. However, their wide employment is hindered by their rapid corrosion behavior. Protective coatings provide a potential approach to extending the service period, but damage to these coatings often leads to local corrosion and even premature failure. To address this issue, self-healing coatings have been developed for providing long-term and reliable protection, even in the presence of defects. This paper summarizes recent progress in self-healing coatings on Mg alloys, with a focus on their uni- and multi-stimuli responsive mechanisms. A typical self-healing coating is composed of a physical layer, inhibitors, and inhibitor containers. Herein, the loading and release of inhibitors are crucial for the design of self-healing coatings. On the one hand, inhibitors can be directly doped/filled into the protective layer and released in response to environmental changes and coating degradation. On the other hand, inhibitors may be encapsulated into micro/nano-containers and released upon being triggered by ions, pH, light, heat, potential and moisture. Additionally, this review presents advanced characterization techniques and systematic evaluation methods for assessing self-healing functionality. Ultimately, the emerging challenges and research priorities in the development of self-healing coatings for Mg alloys are comprehensively discussed.
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1. Corrosion of Mg

Magnesium (Mg) and its alloys are the lightest structural metals, and their high specific strength has attracted significant research interest for lightweight applications. However, Mg exhibits a low standard electrode potential of −2.37 V *vs.* Standard Hydrogen Electrode, rendering it prone to corroding, especially in aqueous media. Paradoxically, the corrosion tendency endows Mg and its alloys with excellent potential as biodegradable metals. The corrosion process of Mg in aqueous media is displayed in Eqs. (1), (2), and (3) [1]. Notably, the naturally formed product layer MgO is porous with a Pilling-Bedworth ratio of <1, which is incapable of protecting Mg substrates. Additionally, the corrosion behavior of Mg and its alloys is also affected by intrinsic and external factors.

$$\text{Anodic reaction: } Mg \rightarrow Mg^{2+} + 2e^- \quad (1)$$

$$\text{Cathodic reaction: } 2H_2O + 2e^- \rightarrow 2OH^- + H_2 \uparrow \quad (2)$$

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$$\text{Total reaction}: Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2 \uparrow \rightarrow Mg^{2+} + 2OH^- + H_2 \uparrow \quad (3)$$

1.1. Intrinsic factors to Mg corrosion

The microstructure is closely related to corrosion behavior [2]. Gusieva et al. [3] reviewed the effects of alloying elements on the corrosion resistance of Mg alloys. The addition of alloying elements affected the anodic or cathodic reaction kinetics of Mg alloys, thereby changing the corrosion current density (i_{corr}) or corrosion potential (E_{corr}). Alloying is a promising approach to improving the corrosion resistance of Mg alloys, such as Al, Zn, Ca, Li, Re, and Mn. Taking Mn as an example, adding Mn content below its solid solubility limit into Mg substrates suppresses the anodic dissolutions without obvious effects on the cathodic processes, therefore improving the corrosion resistance of the Mg alloys. Conversely, Mn contents exceeding the solid solubility limit accelerate the cathodic reaction, deteriorating the corrosion resistance of Mg alloys. Additionally, the standard electrode potentials of impurities (Fe, Cu, and Ni) are significantly higher than the Mg matrix, and these impurities remain insoluble within the Mg matrix, causing microgalvanic corrosion and accelerating substrate degradation.

In addition to alloy elements in Mg alloys, processing technologies, including extrusion, rolling, casting, forging, and drawing, are also employed to regulate microstructures in Mg alloys, such as grain size, second-phase distribution, and crystallographic textures, playing crucial roles in modulating corrosion behavior [4–9]. Besides, heat treatment is also an effective strategy to optimize corrosion resistance through the controlled solid solution of intermetallic compounds and precise reprecipitation of fine second phases [10,11].

1.2. External factors to Mg corrosion

1.2.1. Corrosive solution

To evaluate the corrosion behavior of Mg and its alloys, the selection of appropriate simulated service environments, such as industrial media or physiological solutions, is paramount and should align with specific operational conditions and performance requirements. In industrial applications, Mg alloys encounter aggressive seawater environments, wherein 3.5% NaCl solution is commonly utilized as the simulated corrosive medium. In NaCl solution, the corrosion product is mainly $Mg(OH)_2$, which subsequently converts into soluble $MgCl_2$, and as the dissolution of $MgCl_2$, the corrosion of Mg substrate continues, as shown in Eq. (4) [12]:

$$Mg(OH)_2 + 2Cl^- \rightarrow MgCl_2 + 2OH^- \rightarrow Mg^{2+} + 2Cl^- + 2OH^- \quad (4)$$

Besides, the corrosive characteristic of Mg alloys imparts biodegradation, enabling their application in biodegradable medical implants, whilst the evaluation of the biodegradation behavior of Mg alloys remains a research hotspot. Mei et al. [13] systematically reviewed the current simulated physiological solutions for evaluating the corrosion resistance of degradable medical metals. From NaCl solution to protein-containing cell culture medium, with the addition of inorganic salt ions, small molecular organics, and proteins, the composition of the simulated solution becomes more and more complicated, approximating in vivo physiological conditions.

Notably, NaCl solution is also employed as a simulated physiological medium, which not only isolates the influence of organic and inorganic components on Mg corrosion in physiological solution, but also reflects the inherent corrosion resistance of Mg substrates and coatings. With the addition of inorganic ions, a representative corrosion medium is Hanks' buffered saline solution (HBSS) containing corrosion-mitigating ions such as Ca^{2+}, PO_4^{3-}, and CO_3^{2-}, whose chemical composition is close to the inorganic components of real physiological solution. The inorganic ions (Ca^{2+}, PO_4^{3-}, and CO_3^{2-}) in HBSS solution contribute to the formation of $Mg_mCa_n(PO_4)_x(CO_3)_y(OH)_z$ with good corrosion resistance [14–16]. Furthermore, the cell culture medium, used for in vitro biological evaluation, contains small molecular organics such as amino acids that are absent in HBSS solution, whose inorganic composition predominantly governs Mg corrosion dynamics [17]. Although the composition of the protein-containing cell culture medium is closest to the real physiological solution, the effect of protein on Mg corrosion is still controversial [18,19].

1.2.2. Stress-assisted corrosion

As Mg alloys are increasingly applied in industrial structural parts and medical implant devices, the failure incidences due to stress-assisted corrosion escalate under harsh service conditions [20]. Corrosion fatigue (CF, under cyclic loading) and stress corrosion cracking (SCC, under tensile loading) are typical stress-assisted corrosion behaviors in Mg alloys under the synergistic action of mechanical loading and corrosive medium [21–23]. Mg-based medical implants are subjected to dynamic loads during daily physiological activities [24–26]. The dynamic mechanical loading coupled with the corrosive environment causes significant threats to CF and SCC [27,28], becoming a critical challenge for Mg alloys in industrial and biomedical applications.

Notably, the most fundamental and detrimental feature of CF and SCC is that ductile materials with considerable elongation suffer premature embrittlement in the corrosive environment [29]. The threshold stress for SCC in Mg alloys is far lower than their yield strength, with SCC occurrences even under stresses <60% of yield strength [30,31].

2. Surface modifications and failure modes

2.1. Surface modification strategies on Mg alloys

Surface treatment technologies, as low-cost and efficient approaches, are widely employed to improve the corrosion resistance of Mg alloys. In recent years, the theoretical research and application of surface protection technologies for

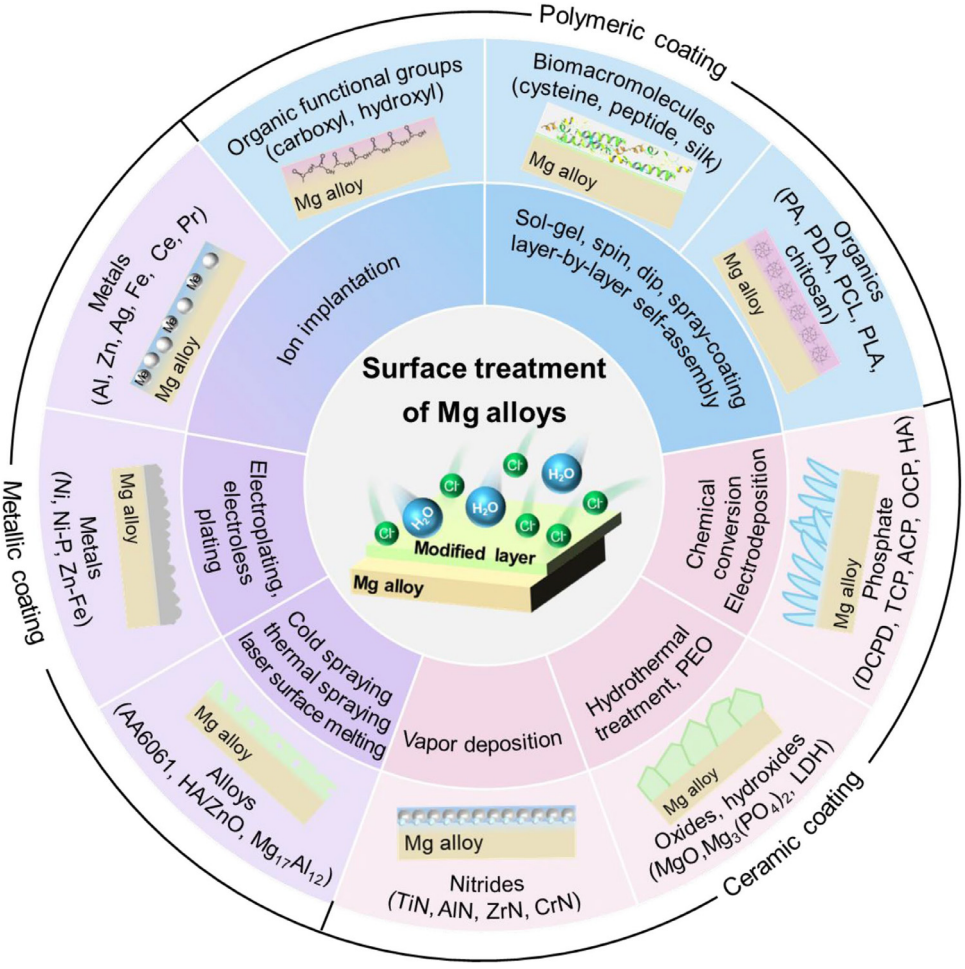


Fig. 1. Surface modifications of Mg alloys.

Mg alloys have been progressively advanced. Fig. 1 presents the typical surface modifications of Mg alloys. Surface modification adjusts surface structure and physicochemical properties by altering surface chemical composition, such as chemical conversion [32–38], micro-arc oxidation (MAO) [39–47], ion implantation [48–56], laser surface nanocrystallization [57,58], hydrothermal treatment [59–63], sol-gel [64,65], physical coating (dip, spin, and spray coating) [66–70], layer-by-layer assembly [71–74], electroplating [75,76], electroless plating [76–79], electrodeposition [80–84], vapor deposition [85–89], thermal/cold spraying [90,91] and laser surface modification [92–95]. Table 1 summarizes surface treatment technologies for Mg alloys.

2.2. Failure forms of coated Mg alloys

The corrosion behavior of surface-treated Mg alloys mainly depends on the surface-modified layers, but when the corrosive solution penetrates to the coating-substrate interface, local pitting corrosion of Mg alloys coupled with H_2 evolution-driven coating delamination is the main corrosive phenomenon. Fischerauer et al. [96] compared the degradation behaviors of MAO-coated and uncoated Mg alloys in rats. The MAO-coated Mg alloys had good initial corrosion resistance in implanted animals, but severe local corrosion occurred in the later stage with faster degradation rates, even exceeding uncoated Mg alloys. Therefore, the evaluation of the corrosion protection performance of coatings on Mg alloys should encompass both the initial corrosion resistance and the long-term protective stability. Fig. 2 is a schematic diagram of the failure behaviors of coatings on Mg alloys in corrosive media [97].

The protective efficacies of metal or conductive ceramic coatings depend on the potential difference between coatings and Mg alloy substrates. A dense and intact coating effectively diminishes the corrosion susceptibility of the Mg alloy substrate. However, once aggressive electrolytes penetrate through coating defects to the coating-substrate interface, galvanic corrosion initiates due to the obvious coating-substrate potential difference, accelerating substrate corrosion and coating failure. While ceramic and polymer coatings can avoid galvanic corrosion due to their insulating nature, it is inevitable to form micropores and cracks in the preparation process. Once Mg alloy degradation occurs, the hydrogen accumulation at the coating-substrate interface will accelerate defect expansion and coating delamination. Subsequent

Table 1
Surface treatment technologies for Mg alloys.

Methods	Phases/Compounds	Advantages	Disadvantages	Ref.
Chemical Conversion	Ca-P (DCPD, TCP, ACP, OCP), MgF_2; Phosphate, molybdate, vanadate, rare earth, PA	➢ Wide application scope; ➢ Simple process; ➢ Low cost.	✧ Small film thickness with poor binding force; ✧ Poor long-term corrosion resistance; ✧ Generally used as transition layers or bottom layers; ✧ Difficult waste liquid treatment.	[32–38]
Micro-arc Oxidation	MgO, $MgAl_2O_4$, Mg_2SiO_4, $MgSiO_3$, $Mg_3(PO_4)_2$, CaP	➢ In-situ grow stable and hard ceramic coating; ➢ Micropores at the micro- and nano-scale for post-treatment; ➢ Strong adhesion strength.	✧ Coating with brittleness and insulation; ✧ High energy consumption; ✧ Necessary post-treatment.	[39–47]
Ion Implantation	Al, Zn, Ag, Fe, Y, Nd, Ce, Pr, carboxyl, hydroxyl, N	➢ Strong adhesion strength; ➢ No obvious interface.	✧ Demanding equipment and high cost; ✧ Unsuitable for complex parts; ✧ Prone to galvanic corrosion.	[48–56]
Surface Nanocrystallization	——	➢ Eco-friendly; ➢ Easy operation.	✧ High cost.	[57,58]
Hydrothermal Treatment	β-TCP/$Mg(OH)_2$, Mg-Al LDH	➢ Economic and energy-saving; ➢ Easy operation.	✧ High process temperature and pressure.	[59–63]
Sol-gel Method	HA, silane-based alkolsol	➢ Simple preparation process with low reaction temperature; ➢ Organic-inorganic hybridization and diverse functionalities.	✧ Essential liquid compound precursor; ✧ Unavoidable pores and cracks from drying and curing; ✧ Only suitable for high-temperature-resistant substrates.	[64,65]
Physical Coating (Dip, Spin, and Spray Coating)	PLA, PDA, PTMC, chitosan, PBA, PDLLA	➢ High efficiency and easy operation; ➢ Large-area and complex-shape adaptability (as for spray-coating).	✧ Poor thickness control; ✧ Solvent defects; ✧ Splashing and edge effects (as for spin and spray-coating).	[66–70]
Layer-by-layer Self-assembly	PA, PEI/PSS/8-HQ, PDA/Ag, Poly-L-lysine/alginate	➢ Controllable molecular-level thickness; ➢ Multi-functional loading.	✧ Low efficiency and time-consuming; ✧ Relatively low mechanical strength.	[71–74]
Electroplating	Ni, Zn-Fe	➢ Low cost.	✧ High energy consumption; ✧ Difficult to recycle electrolyte; ✧ Nonuniform film.	[75,76]
Electroless Plating	Ni, Ni-P	➢ Low energy consumption.	✧ Unstable service life of plating bath; ✧ Difficult to recycle plating solution; ✧ Nonuniform film.	[76–79]
Electrodeposition	DCPD, HA, $Mg(OH)_2$, Magnesium nitrate, Cu/Ni-P	➢ Available metal coating; ➢ Short deposition time.	✧ High energy consumption; ✧ Difficult to recycle electrolytes.	[80–84]
Vapor Deposition	TiN, AlN, ZrN, CrN, TiMgN, TiMgGdN, TiMgYN	➢ Uniform and dense film; ➢ Strong adhesion strength.	✧ Complicated equipment and high cost.	[85–89]
Thermal/Cold Spraying	HA/ZnO, AA6061	➢ High deposition efficiency; ➢ Strong adaptability.	✧ High energy-consuming and equipment cost; ✧ Residual stress.	[90,91]
Laser Surface Modification	$Mg_{17}Al_{12}$, Amorphous phase	➢ Rapid processing speed; ➢ High-precision microstructural control.	✧ High requirements of equipment and cost; ✧ Residual stress; ✧ Limited substrate compatibility.	[92–95]

Abbreviations: Dicalcium phosphate dehydrate (DCPD), Amorphous calcium phosphate (ACP), Octacalcium phosphate (OCP), Tricalcium phosphate (TCP), Phytic acid (PA), Hydroxyapatite (HA), Polylactic acid (PLA), Polydopamine (PDA), Poly (trimethylene carbonate) (PTMC), Poly(butyl acrylate) (PBA), Poly(d,l-lactide) (PDLLA), Poly(ethylene imine) (PEI), Poly(styrene sulfonate) (PSS), 8-hydroxyquinoline (8-HQ).

ingress of more corrosive ions through coating defects ultimately results in the failure of the coating. Besides, under a physiological mechanical-chemical coupling environment, stress also causes the coating crack/delamination, leading to the risk of localized corrosion and premature failure of coated Mg alloys.

Although single-layer coatings (metallic, ceramic, and polymer coatings) struggle to provide long-term corrosion protection for Mg alloy substrates, multilayer coating emerges as a promising alternative strategy. The combination of pretreatment and post-processing could obtain protective coatings well-bonded to the substrates [98–101].

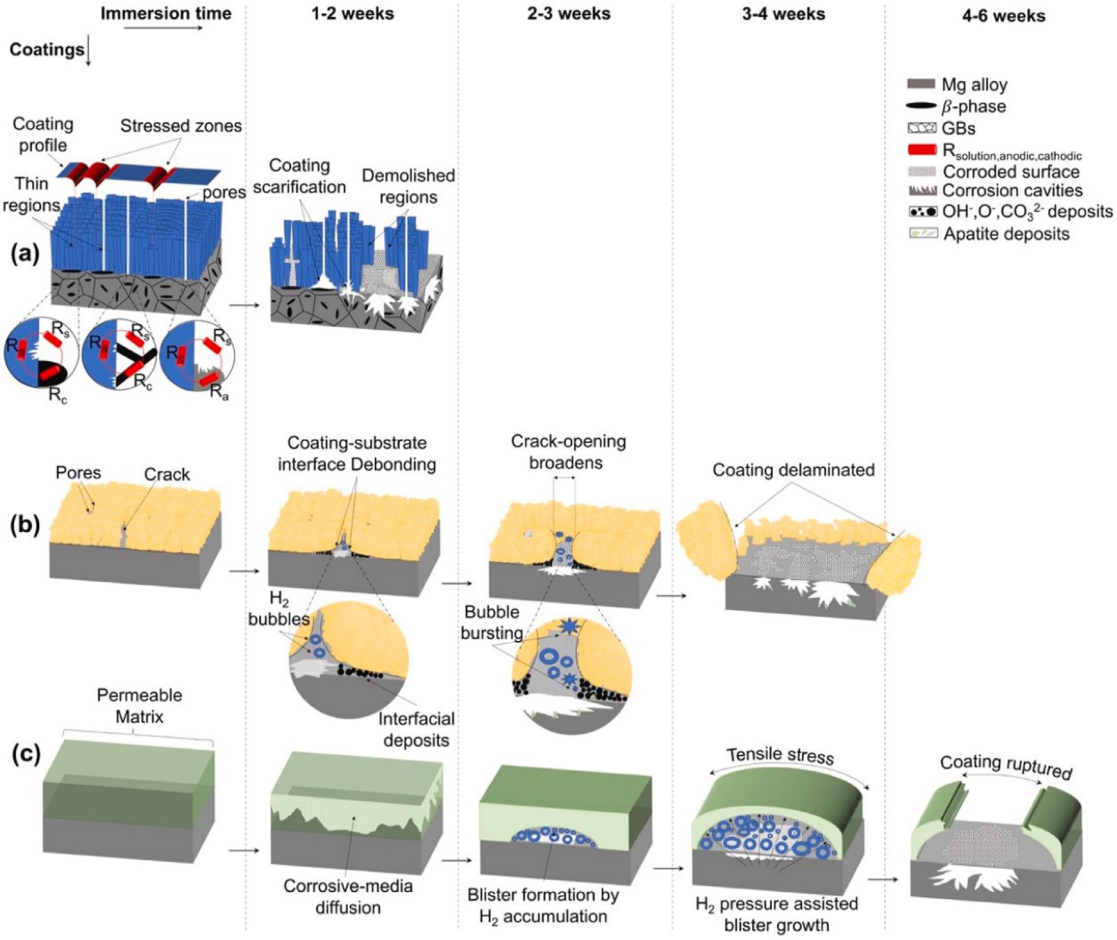


Fig. 2. Schematic illustration of immersion-time-based failure behaviors of (a) metallic/conductive coatings, (b) ceramic coatings, and (c) polymer coatings on Mg alloys in physiological media [97].

3. Self-healing coatings

Surface treatment is an effective method of mitigating corrosion for metallic materials. While conventional coatings provide "passive" protection as corrosion barriers, their inherent limitations in long-term stable protection become critically evident under dynamic service conditions. Especially in the complex service environment, the inevitable mechanical and/or chemical damages initiate local corrosion, which subsequently propagates along the interface to compromise coating adhesion, leading to coating delamination and failure eventually.

A concept of smart coatings has been proposed and developed to promote surface coatings towards intelligence [102]. Fig. 3 outlines the design ideas for smart self-healing coatings. The self-healing coatings are sensitive to internal and/or external stimuli, including light, heat, moisture, ΔpH, and electric potential (ΔΦ). Afterward, the self-healing coatings actively resist detrimental effects and rapidly restore the integrity to maintain long-term corrosion resistance and structural durability. Self-healing functionality is typically realized through either the inherent self-healing capacity of the protective coating itself or the synergistic action of embedded corrosion inhibitors. Corrosion inhibitors play a crucial role in the repair process for inhibitor-loaded coatings, which are usually added to the coatings by direct doping or container encapsulation. Upon damage to the coating or activation of stimuli, inhibitors are released and either interact with substrate-derived corrosion products or adhere directly via electrostatic attraction and chemical binding to seal the defects.

Hence, it is prospective and urgent to develop multifunctional coatings that integrate self-healing capability, environmental adaptability, and structural durability to extend the applications scope of Mg alloys, particularly in aerospace, automotive, and biomedical industries.

According to the structural design, self-healing coatings can be systematically categorized into three primary types, including self-healing protective coatings, inhibitor-loaded self-healing coatings, and self-healing coatings incorporated with inhibitor-loaded micro/nano containers. Fig. 3(A-C) are schematic diagrams of the structural design of three types of self-healing coatings. The protective coatings with self-healing function act as both physical barriers against corrosive media and corrosion inhibitors for defect repair. The inhibitor-loaded self-healing coatings are constructed by doping or filling the corrosion inhibitors into the protective coating.

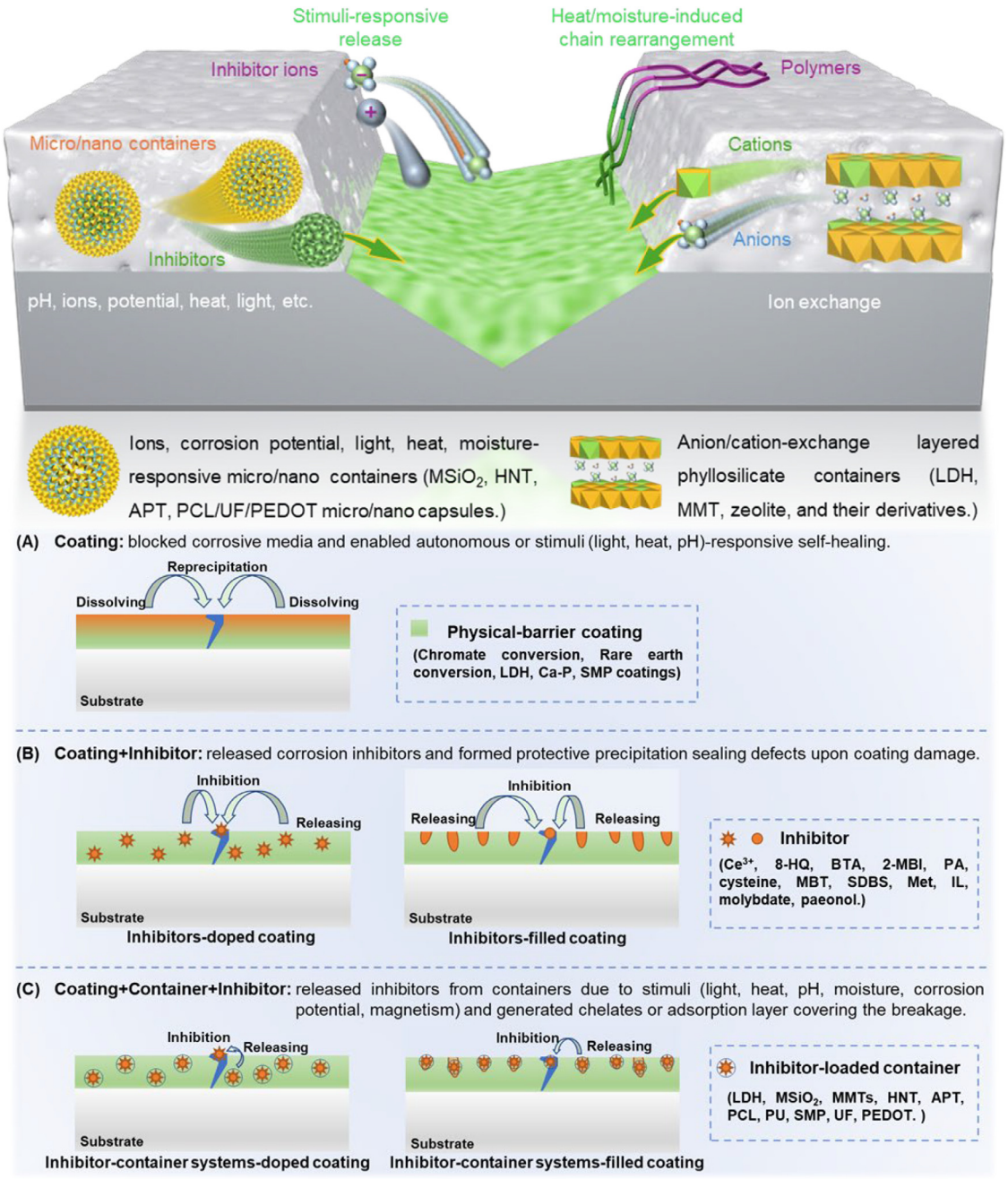


Fig. 3. The schematic diagram of self-healing coatings for corrosion protection and three representative categories of these coatings: (A) self-healing protective coatings; (B) corrosion inhibitor-loaded self-healing coatings; (C) self-healing coatings incorporated with corrosion inhibitor-loaded micro/nano containers.

The self-healing coatings incorporated with inhibitor-loaded micro/nano containers include physical layers, corrosion inhibitors, and containers. In coating design, the continuous innovation of multi-component integration and smart additive incorporation is gradually driving the evolution of self-healing coatings towards diversification, multi-functionalization, and intelligence.

In industrial applications, self-healing coatings primarily focus on long-term protective properties, in which anti-corrosive physical layers and highly efficient corrosion inhibitors are used for coating design. Nevertheless, biomedical self-healing coatings must consider biosecurity and provide suitable corrosion protection for temporary Mg-based implants. Also, biological functions, such as antibacterial and osteogenic properties, are recommended to be introduced into biomedical self-healing coatings for efficient tissue repair. Meanwhile, stimulus-responsive approaches to achieving self-healing have limitations and require adaptation to match the physiological balance, including physiological pH and temperature range.

Table 2
Spontaneous self-healing coatings on Mg alloys.

Coating	Substrate	Mechanism	Ref.
Chromate conversion coating	AZ31B Mg alloy	The Cr-based oxides or compounds formed and deposited on the defects.	[103]
Stannate conversion coating	AZ91D Mg alloy	The $Mg(OH)_2$ layer enriched with tin-oxide was formed to repair the surface defects. The $MgSnO_3$ layer dissolved and reprecipitated on the surface.	[104,105]
Vanadate conversion coating	AZ31 Mg alloy	The vanadium ions were released from coatings and migrated to the damaged zone to form a new layer.	[36,106]
Rare earth conversion coating	AZ80X Mg alloy Mg-Nd-Gd-Zn (EV31) alloy WE43 Mg alloy AZ31B Mg alloy	An insoluble passive rare earth oxide/hydroxide film was deposited at cathodic domains due to the alkaline pH.	[107,108]
Phosphate-based PEO coating	ACM522 Mg alloy AZ31 alloy and pure Mg	The PO_4^{3-} in the coating formed stable product films with Mg^{2+} at scratches.	[109]
Al-alloyed surface	AZ31 Mg alloy	The crystallization of LDHs occurred inside the defect zones during immersion.	[110]
MnOOH	PEO-coated Mg	LDHs were formed in the scratches rather than $Mg(OH)_2$.	[12]
HA/OCP	Pure Mg and AZ31	The pH increase and released phosphate result in the precipitation of Mg, Ca, and P-containing compounds.	[111]
ACP	Mg-2.2Zn-0.3Ca alloy	Ca^{2+} and PO_4^{3-} ions were released from ACP due to local alkalinity and reacted to form a corrosion-resistant product layer to heal defects.	[112]
DCPD	Pure Mg	Localized alkalinity triggered the release of Ca^{2+} and PO_4^{3-} ions from DCPD, and Ca, P-containing compounds were formed for self-healing.	[113]

3.1. Protective coatings with self-healing function

3.1.1. Spontaneous self-healing coatings

Coatings with self-healing ability are accompanied by the dissolution and redeposition of inorganic components from the coating in corrosive solutions, thereby restoring coating integrity and mitigating matrix corrosion. Generally, this category predominantly includes conversion coatings prepared by chemical conversion and plasma electrolytic oxidation (PEO), such as cerium (Ce)-, chromate-, phosphate-, and silicate-based coatings. Spontaneous self-healing coatings on Mg alloys are summarized in Table 2.

The hexavalent chromium (Cr^{6+})-based conversion coatings on Mg alloy substrates demonstrated the characteristic self-healing function. Once Cr^{6+}-containing coatings were damaged, the released Cr^{6+} ions underwent redox reactions to form a Cr^{3+}-rich chromate coating at the broken regions (Fig. 4(B)) [103]. Nevertheless, due to the environmental pollution and carcinogenicity of Cr^{6+}, chromate-based coatings have been strictly restricted by the European Union (EU) Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) [114,115]. Therefore, there is an urgent requirement to develop eco-friendly coatings with excellent corrosion resistance and self-healing function to replace chromate conversion coatings. Notably, for biomedical implants, good biocompatibility must be strategically incorporated into the coating designs, except for corrosion mitigation and self-healing capacity.

Chromium-free eco-friendly coatings, stannate [104,105], vanadate [36,106], and rare earth conversion coatings [107,108] on Mg alloys exhibited self-healing functions through the precipitation of metal oxides and hydroxides in the damaged area. As for rare earth conversion coatings, the Ce-containing oxides/hydroxides precipitated in active anodic sites, restraining local corrosion through passive precipitation [116]. Furthermore, Ce ions also exhibit synergistic compatibility with additional corrosion inhibitors, facilitating the integration of cerium salt conversion coatings into self-healing coating systems design [117]. Jian et al. [107] developed Mn-Ce hybrid coatings on EV31 Mg alloys and demonstrated that Mn^{7+} served as a redox mediator to facilitate the oxidation of Ce^{3+} to Ce^{4+}, thereby depositing Ce^{4+} oxide/hydroxide layers in the cracked zones (Fig. 4(A)). Except for Ce-based coatings, the praseodymium (Pr) conversion film on Mg alloy also spontaneously precipitated Pr oxide/hydroxides to cover defects [108]. It is worth mentioning that rare earth elements are non-essential trace elements for human physiology and have potential adverse effects on human metabolism, such as long-term chronic toxicity caused by element accumulation [118,119]. Therefore, rare earth conversion coatings are unsuitable for biomedical Mg alloy implants due to insufficient biosafety.

Phosphate ion (PO_4^{3-}), as an inorganic salt composition in cells, plays a vital role in cell metabolism, such as buffering pH, energy transfer, which meet the biosafe and biocompatible requirements of medical Mg alloys. Williams et al. [120] investigated the inhibitory effects of Ce^{3+}, Y^{3+}, F^-, CrO_4^{2-}, and PO_4^{3-} ions on the localized corrosion behavior of AZ31 Mg alloys, with PO_4^{3-} and CrO_4^{2-} ions exhibiting notable efficacy in corrosion inhibition. It was reported that phosphate- and silicate-based PEO coatings on Mg alloys showed self-healing ability due to PO_4^{3-}/amorphous SiO_2 dissolution and Mg^{2+}-complexed deposition on scratched surfaces (Fig. 4(C)) [109]. However, large pore defects compromised coating self-healing capacity, as the formed products fail to seal the defects during immersion corrosion [121]. Moreover, PO_4^{3-} ion is also an essential component in synthesizing Ca-P compounds, including HA, OCP, ACP and DCPD. Hiromoto [111]

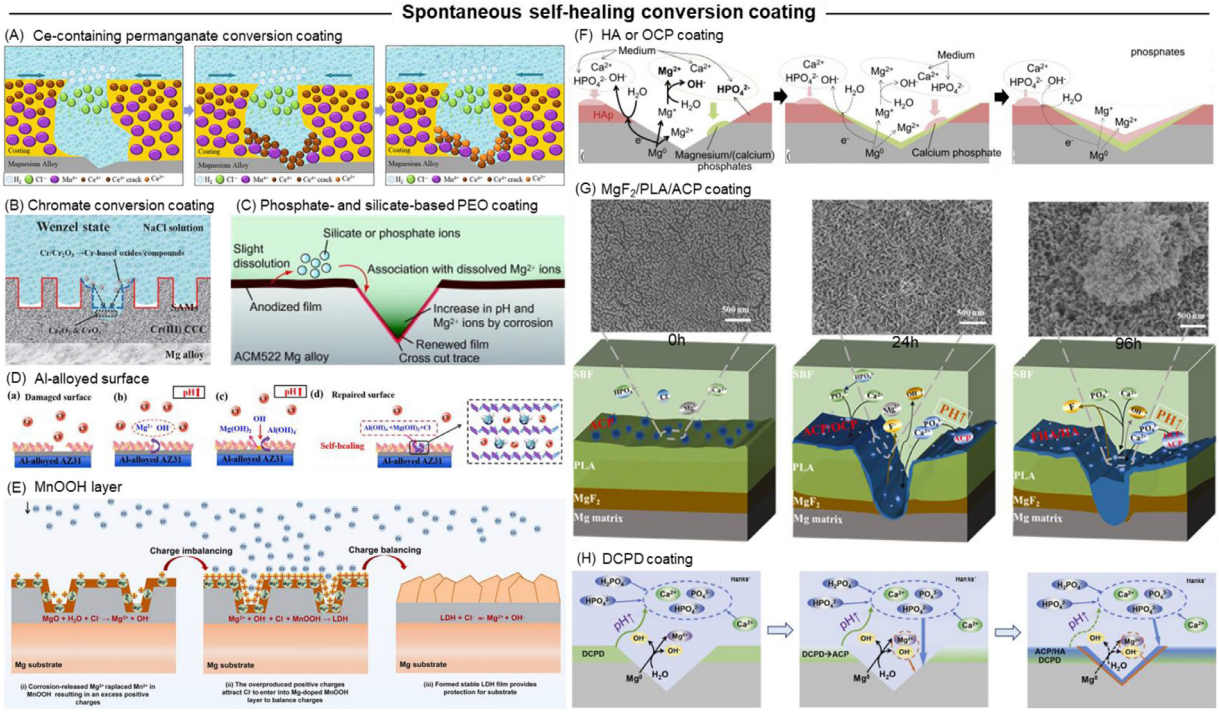


Fig. 4. Schematic illustrations of the spontaneous self-healing process of conversion coatings on Mg alloys: (A) the cerium-containing permanganate conversion coating [107]; (B) the chromate conversion coating [103]; (C) the phosphate- and silicate-based PEO coatings [109]; (D) Al-alloyed AZ31 [110]; (E) the MnOOH film on PEO-coated Mg [12]; (F) the HA or OCP coating [111]; (G) the MgF_2/PLA/ACP coating [112]; (H) the DCPD coating in HBSS solution [113].

found that the pH increase at scratched regions and the release of PO_4^{3-} ions from the HA and OCP coatings promote the deposition of $Mg_3(PO_4)_2$/$Mg(OH)_2$ and $Ca_3(PO_4)_2$ protective product layers, sealing the scratches and mitigating corrosion. These findings have greatly advanced the development of inspired self-healing coatings in biomedical applications. Liu et al. [112] deposited ACP film on MgF_2/PLA coating out of consideration for the pH-responsive characteristic of ACP. The transformation from ACP to OCP and HA was attributed to acidic microenvironments from PLA hydrolysis and local alkaline pH owing to Mg corrosion, respectively. Among Ca-P compounds, DCPD has poor chemical stability and is regarded as a precursor of stable HA and OCP, therefore, it has the remarkable potential for enabling pH-induced self-healing of coatings [122]. DCPD coatings with high calcium-phosphorus content and pH-responsive capacity tended to dissolve and release Ca^{2+} and PO_4^{3-} ions quickly to form stable precipitations in local alkaline corrosion regions [113]. Fig. 4(F-H) are the schematic diagram of the self-healing mechanisms of HA, ACP, OCP, and DCPD coatings on Mg alloys. By designing and controlling pH values in microenvironments, pH-dependent Ca-P compounds (ACP, DCPD, OCP, and HA) undergo dynamic phase transitions, enabling reversible "dissolution-deposition" cycles in self-healing coatings. Recently, DCPD coating was used for Mg alloy implants to overcome coating damage from deformation as anastomotic staples [123].

The "dissolution-recrystallization" process of layered double hydroxide (LDH) also enables LDH coating with self-healing capacity, even without loading corrosion inhibitors, benefiting from intrinsic structure reconstruction. Local alkalization and Mg^{2+} ions accumulation from the Mg corrosion process contribute to LDH formation derived from extra trivalent cations in the coatings and substrates. When Mg^{2+} and Al^{3+} ions reach the critical solubility product, heterogeneous nucleation is triggered, and recrystallization occurs via ion exchange. Fig. 4(D) exhibits the schematic diagram of the growth-related self-healing mechanism of LDH conversion films on the Al-alloyed AZ31 [110]. Al-alloyed surface in the scratched region released Mg^{2+} and $Al(OH)_4^-$ ions, and these ions combined with OH^- and Cl^- to grow Mg-Al LDH–Cl^- film, blocking local corrosion and realizing self-healing [124]. Similarly, a self-healing coating generated from the corrosion reaction was constructed via PEO/MnOOH films on Mg alloys. In Fig. 4(E), the dissolved Mg^{2+} diffused into the MnOOH interlayers to replace Mn^{3+} spontaneously, subsequently the overproduced positive charges attracted anions like Cl^- and OH^-, forming charge-balanced LDH, providing much better corrosion resistance performance compared with $Mg(OH)_2$ [12].

In summary, the spontaneous self-healing process of the protective coating depends on the dissolution and redeposition of active ingredients in the coating in corrosive media. The dissolved ions from the coatings diffuse towards defects

and microcracks to form a stable protective layer. In the protective coating with self-healing function, the coating not only acts as a protective barrier against the corrosive media, but also spontaneously initiates a "dissolution-deposition" cycle for sealing cracks and providing long-term protection.

3.1.2. Inspired self-healing coatings

Although conventional protective coatings exhibit limited spontaneous defect repair or coating recovery, the inspired coatings have been developed with reasonable compositions to gift themselves self-healing capacities of responding to external stimuli (light, heat, and moisture). Generally, the self-healing mechanisms of external stimuli-inspired coatings on Mg alloys mainly involve intrinsic light, heat, and moisture-induced reversible chemical/physical interactions of polymers. These mechanisms enable coatings to dynamically adapt to corrosive environments through phase reconstitution or reversible bond rearrangement, thereby maintaining long-term structural integrity and corrosion mitigation.

It is well-known that intrinsic self-healing polymers possess diverse reversible bonds, including hydrogen bonds, imine bonds, coordination bonds, disulfide bonds, and boronate ester bonds. They can restore original structure autonomously through external stimuli (light, heat, and moisture)-assisted chemical/physical interactions [125]. Shape-memory polymer (SMP) coatings have emerged as promising candidates for self-healing systems, enabling a transition from barrier protection to active repair due to the shape-memory effect activated through electrothermal or photothermal stimuli. Fig. 5 illustrates the self-healing mechanism diagrams of SMP coatings. Zhang et al. [126] constructed a self-healing epoxy resin (SHEP) coating possessing a thermally actuated shape memory effect, followed by a porous superamphiphobic perfluorodecyl polysiloxane (PF-POS) modified silica coating. The heat-activated shape memory effect enabled SHEP coating to recover original structure and further drove the healing of the upper superamphiphobic coating, blocking the ingress and diffusion of corrosive species. To avoid relying on external thermal stimuli, photothermal conversion dopants, such as multiwalled carbon nanotubes (MWCNTs) [127] and carbon nanospheres [128], were incorporated into the SMP matrix. Under natural light, spontaneous repair of SMP coatings is activated to restore their original shape through chain untangling and rearrangement. Studies have reported that heat-activated self-healing polymers in epoxy resin coating spontaneously diffuse towards and cover up the defects driven by lower surface energy [129,130]. Polycaprolactone (PCL), as a well-known thermoresponsive SMP with a low glass transition temperature, is frequently incorporated into self-healing coatings. Yang et al. [131] prepared MIL-53@TA/PCL coatings on AZ31 Mg alloys by using PCL as resin matrix and tannic acid (TA) modified iron-based metal-organic framework MIL-53 as functional filler with excellent photothermal conversion ability. Under the irradiation of near-infrared light (NIR), the macromolecular chains of heated PCL migrated to the defects. Likewise, the photothermal effect of polypyrrole (PPy) under NIR irradiation prompted the shape deformation of PCL for defect closure [132], as shown in Fig. 5(C). The heat-induced self-healing mechanism for restoring structural integrity relies on reversible crystallization transitions and macromolecular chain rearrangement.

Except for light and heat, moisture, as a common and excellent hydrogen bond donor/acceptor, has been widely applied in self-healing polymer systems through moisture-induced plasticization, hydrolysis, and solidification. The weakened intermolecular hydrogen bonds, the improved molecular mobility, as well as the enhanced dynamicity of crosslinks facilitate the self-healing process [133]. Deng et al. [134] sprayed nitrogen-ligated boroxine cross-linked polydimethylsiloxane (N-B-PDMS) coatings on the Mg-3.0Zn-0.5Ca alloys, featuring reversible covalent boroxine/hydrogen bonding networks and moisture-stimulated self-healing functionality (Fig. 5(D)). Besides, the swelling of layer-by-layer assembly PEI/poly(acrylic acid) (PAA) promoted molecular mobility and defect repair with the assistance of electrostatic attractive forces [135]. Also, Zhang et al. [136] synthesized stable poly(thioctic acid(TA)-Zn) polymer coatings on ZE21B Mg alloys by replacing partial weak hydrogen bonds with Zn-COOH coordination bonds. When hydration-induced dissociation of the Zn-COOH complex occurred, the active repair initiated, including the noncovalent recombination of non-saturated hydrogen bonds and Zn-COOH coordinative bonds, and the reconstruction of polymer network via disulfide dynamic covalent exchange.

While the shape memory effect and swelling of polymers can effectively restore coating integrity, they cannot completely prevent electrolyte penetration into the coating/substrate interface, which results in the dissolution of the substrate and the stripping of the coating eventually. To address the electrolyte penetration-induced substrate corrosion and hydrogen evolution-induced multilayer delamination, three interconnected microporous PCL layers were prepared on the underlying pore-free dense PCL layer over HA/TiO_2-coated samples [137]. The topmost interconnected microporous PCL layers facilitated the escape of hydrogen gas, avoiding interfacial separation and coating failure, and the bottom physical protective layer restricted the Mg dissolution. Moreover, the contraction of PCL micropores due to water uptake reduced the spacing of polar groups within micropores, augmenting Van der Waals forces on account of stronger dipole interaction, therefore facilitating the formation of 3d flower-like HA crystals in simulated body fluid (SBF) solution to fill PCL micropores [138].

3.2. Self-healing coatings constructed by protective layers and corrosion inhibitors

While many coatings inherently possess self-healing functionality, the majority of coatings merely function as protective barriers and lack self-healing properties. The artificial incorporation of corrosion inhibitors is utilized to endow coatings with self-healing properties, thereby extending their service lifespan. In general, a self-healing coating

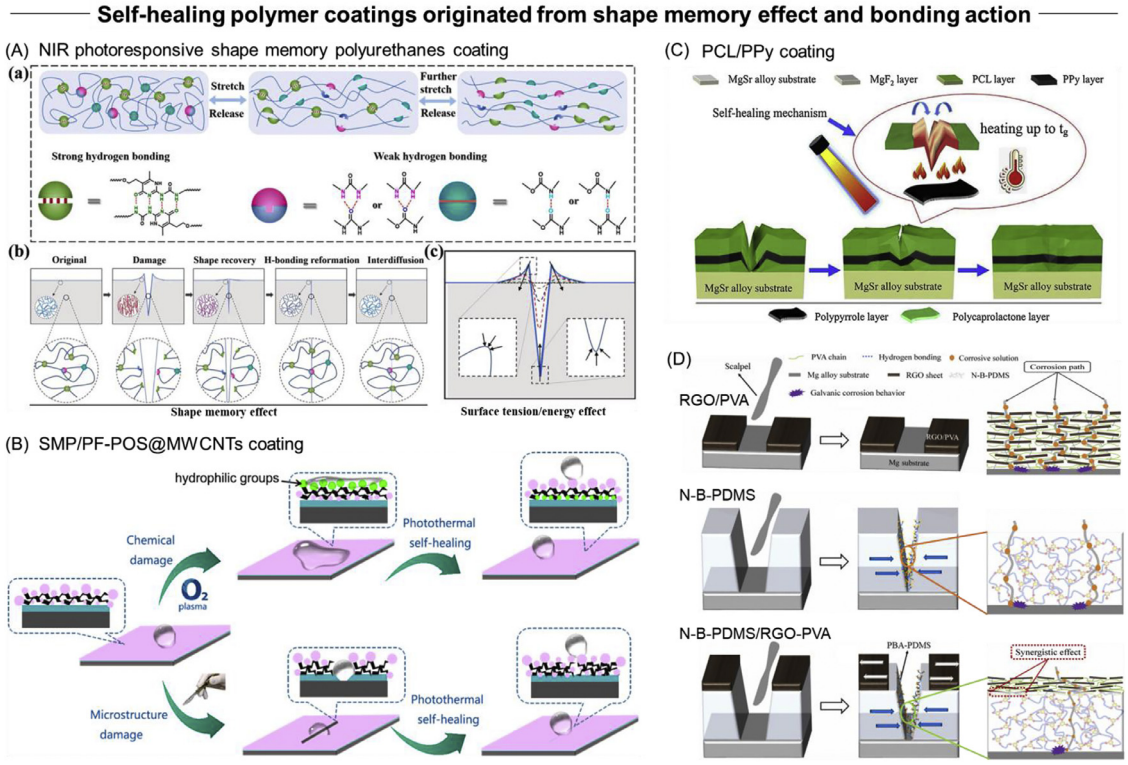


Fig. 5. Self-healing polymer coatings originated from shape memory effect and bonding action: (A) NIR-responsive SMP coatings: (a) hierarchical hydrogen bonding; (b) shape memory effect and (c) the action of energy/tension [128]; (B) SMP/PF-POS@MWCNTs coatings [127]; (C) PCL/PPy composite polymer coatings [132]; (D) the reduced graphene oxide/polyvinyl alcohol (RGO/PVA), N-B-PDMS, and N-B-PDMS/RGO-PVA coatings [134].

consists of a "guest" for corrosion mitigation, and a "host" as a physical barrier and corrosion inhibitor carrier [139]. Doping and filling are two primary strategies for inhibitor incorporation. Doping involves the blending of inhibitors and coating materials before preparation, whereas filling involves the delivery of inhibitors into microscopic pores or defects in the prepared underlying coating.

3.2.1. Inhibitor-doped self-healing coatings

Doping corrosion inhibitors directly into the coatings can impart self-healing capabilities to physical barrier coatings. Galio et al. [140] doped 8-HQ into the SiO_2-ZrO_2 sol-gel coating, and the stable $Mg(8\text{-}HQ)_2$ compound sealed defects in SiO_2-ZrO_2 coating. Moreover, hydroxyquinoline (HQ) coatings doped with graphene oxide (GO) particles were fabricated on the oxidized/phosphatized AZ31B Mg alloys. $Mg(HQ)_2$ chelates and the swelling and mixing of freely active-charged groups from GO particles in the damaged zone impeded jointly electrolyte diffusion and corrosive ions attack [141]. Ce-based compounds (Ce oxides and hydroxides) exhibit exceptional oxidation-reduction cycling, making them ideal candidates for corrosion inhibitors in self-healing coatings. Polyacrylamide (PAM) gels with Ce^{4+} inhibitors were synthesized on the porous MAO-treated AZ31 Mg alloys, and pH-responsive swelling of PAM gels triggered the release of Ce^{4+}, afterward forming Ce oxides and hydroxides compact protective film [142]. Calado et al. [143] applied CeO_2-modified hybrid epoxy-silane coatings on AZ31 Mg alloys. CeO_2 nanoparticles played multiple roles in adsorbing aggressive species (Cl^-), stabilizing corrosion products, and blocking electrolyte diffusion into coatings.

Given the limited doping concentrations of corrosion inhibitors in the protective coating and the introduction of defects during fabrication, it is particularly critical to improve interfacial compatibility between corrosion inhibitors and coatings and minimize inhibitor waste. To improve compatibility between inorganic inhibitors and organic host matrix, organic anions and the cerium cations were combined to synthesize organic cerium tri(bis(2-ethylhexyl)phosphate) ($Ce(DEHP)_3$), which was doped into the epoxy-silane coating. The dissociation of $Ce(DEHP)_3$ due to local alkalization promoted the formation of $Ce(OH)_3$ and magnesium organophosphate, sealing the defects (Fig. 6(A)) [144,145]. Besides, Xiong et al. [146] prepared K_3PO_4-doped silk fibroin coatings on Mg alloys, wherein the K^+ ions regulated the secondary structure of silk fibroin, promoting a preferable adhesion to the substrate. Silk fibroin responded to high pH and the released PO_4^{3-} ions, forming $Mg_3(PO_4)_2$ precipitates on the defects (Fig. 6(B)).

PA is an organic molecule extracted from plants. It has been employed for Mg alloy surface modification owing to its strong complexation ability toward metal ions, and the resultant complexes cover the defects and restore coating integrity (Fig. 6(C)) [147]. PA/3-aminopropyltrimethoxysilane (γ-APS) coatings were synthesized on AZ31 Mg alloys, and water permeation-induced local pH elevation triggered PA re-

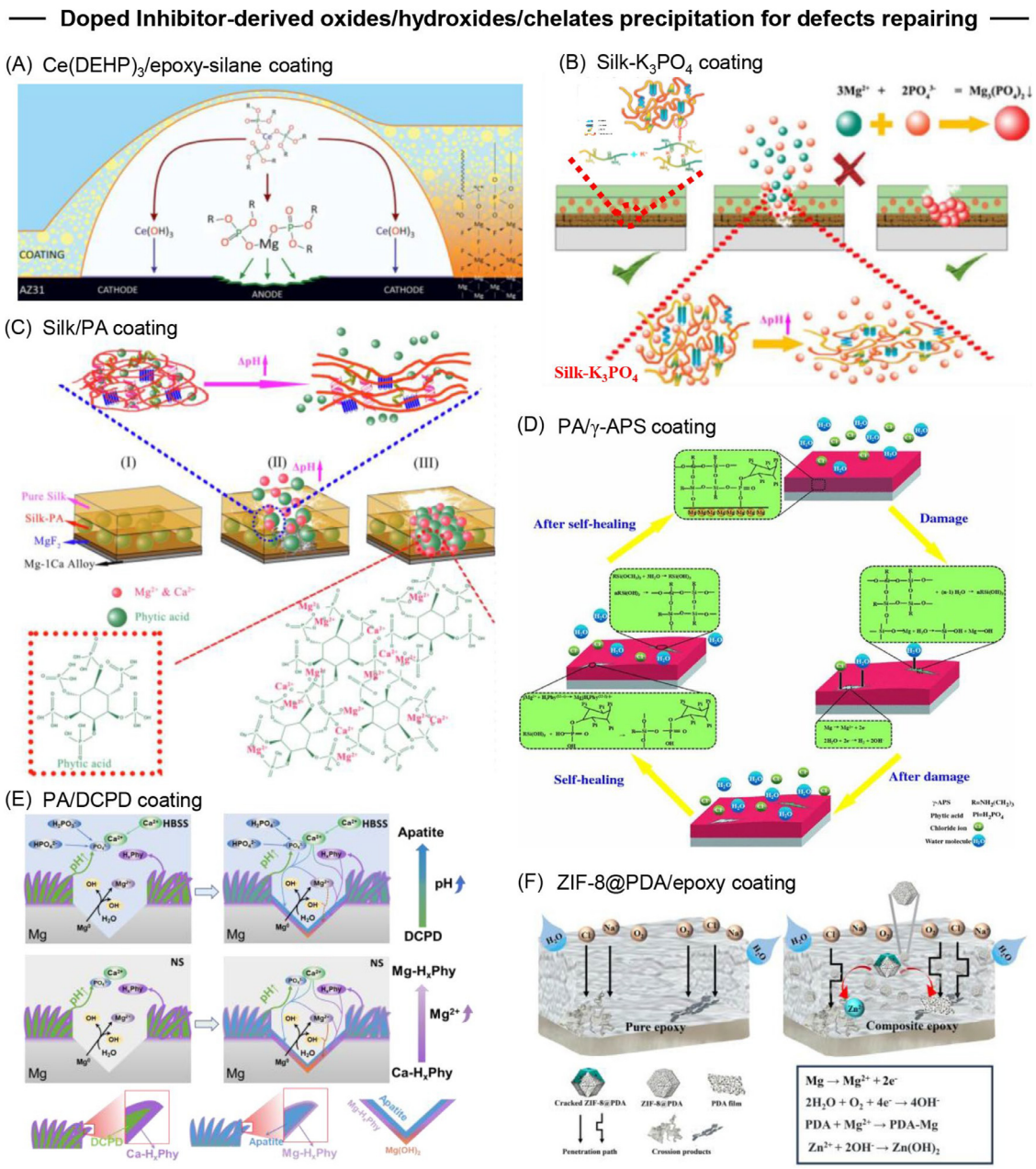


Fig. 6. Self-healing mechanism of corrosion inhibitor-derived oxides/hydroxides precipitation: (A) the $Ce(DEHP)_3$-doped epoxy-silane coating [144]; (B) the Silk/K_3PO_4 coating [146]; (C) the Silk/PA coating [147]; (D) the PA/γ-APS hybrid coating [148]; (E) the PA/DCPD coating [149]; (F) the ZIF-8@PDA-doped epoxy coating [151].

lease from the hybrid coating, followed by Mg^{2+} chelation and precipitation on the defects (Fig. 6(D)) [148]. As shown in Fig. 6(E), the PA/DCPD coating was constructed on the pure Mg surface. Local pH increase contributed to the dissolution of Ca^{2+} and PO_4^{3-} ions from DCPD coating and the formation of corrosion-resistant apatite products. Meanwhile, the Mg^{2+} enrichment caused by Mg corrosion induced Mg-PA deposition on broken regions for isolating further erosion [149].

Except for single corrosion inhibitors, the codoping of multiple inhibitors has been extensively investigated, which imparts multiple self-healing capacities to coatings. The SMP coating codoped with benzotriazole (BTA) and ceresine wax was prepared on AZ31B Mg alloys. Electrolyte permeation induced the release of BTA, which then adsorbed onto the metal surface. Besides, upon thermal activation, the shape memory effect of SMP and the sealing of ceresine wax synergistically repaired defects [150]. The PDA polymer membrane on the ZIF-8 surface improved the dispersion of ZIF-8@PDA nanocapsules in the epoxy matrix and the interfacial compatibility between ZIF-8 and epoxy resin. On the one hand, the pH-responsive PDA membrane separated from ZIF-8 surface and then coordinated with Mg^{2+} ions to generate PDA-Mg complexes on the corroded zones. On the other hand, Zn^{2+}

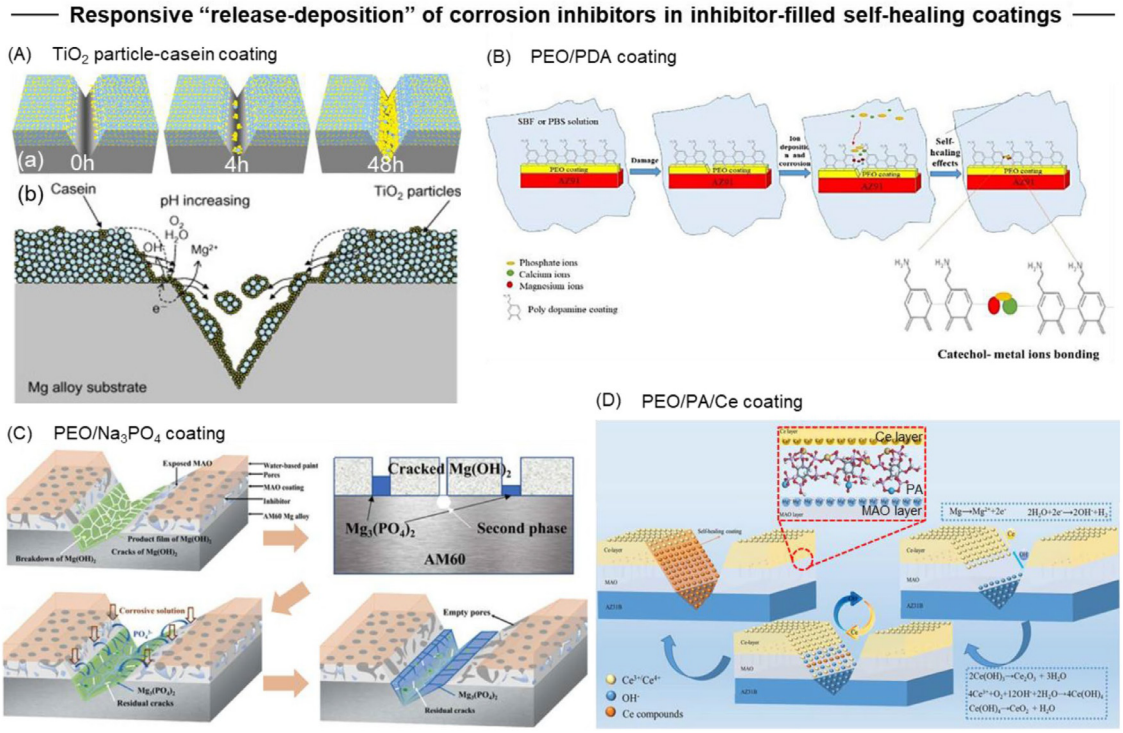


Fig. 7. Self-healing process and mechanism of pH-responsive casein inhibitor-filled TiO_2 particle films (A): (a) the scratched area of TiO_2 particle-casein coating by changing the pH from 12 to 5, (b) the self-healing mechanism [152]; and schematic illustration of "release-deposition" self-healing of inhibitor-filled PEO coatings: (B) the PEO/PDA coating [153]; (C) the PEO/Na_3PO_4 coating [154]; (D) the PEO/PA/Ce coating [156].

released from ZIF-8 converted to $Zn(OH)_2$ precipitates, reinforcing corrosion inhibition and defect closure (Fig. 6(F)) [151].

The preparation process of the inhibitor-doped self-healing coating is relatively simple, whereas the doping forms, varieties, and concentrations of corrosion inhibitors into the coatings should be appropriate for improving the self-healing capability and reducing defect introduction as much as possible synergistically.

3.2.2. *Inhibitor-filled self-healing coatings*

In inhibitor-loaded self-healing coatings, the protective layer not only acts as a physical barrier against corrosion, but also serves as a container of corrosion inhibitors. Additionally, the incorporated inhibitors reduce coating porosity through micropore filling, enhancing densification and sealing defects. Yabuki et al. [152] inserted pH-responsive casein inhibitors into TiO_2 particle films on AZ31 Mg alloys for improving the compactness of TiO_2 layers. Fig. 7(A) shows pH-induced morphology evolution of the scratched area in casein-containing TiO_2 film through immersion durations of 0 h, 4 h, and 48 h (pH adjusted from 12 to 5). TiO_2 particles and casein played complementary roles during the self-healing process: casein was released from the TiO_2 film owing to local pH increase, while in return, TiO_2 particles were transported by casein towards defect zones.

PEO coating is a representative physical barrier layer with the outer porous layer and inner dense layer, which enables inhibitor-loaded capability and isolation effect between Mg matrix and corrosion inhibitors. Farshid et al. [153] electrodeposited PDA on PEO-coated AZ91 Mg alloys. The spontaneous formation of hydrogen bonding between $Mg(OH)_2$ and catechol groups of dopamine, along with the bonding between Ca^{2+}, PO_4^{3-} from SBF solution and catechol over PDA layers synergistically promoted defect closure (Fig. 7(B)). Inorganic coatings have also been developed for filling micropores on PEO coatings [154]. For instance, $CaCO_3$ was filled into PEO coating on AZ41 Mg alloys by ultrasound-assisted chemical conversion. The self-healing process comprised the formation of calcite through the solid-state transformation and the $CaCO_3$ precipitation from $Ca(OH)_2$ in immersion solution [155]. The self-healing mechanism of inorganic inhibitor (Na_3PO_4)-filled PEO coating was attributed to the deposition of $Mg_3(PO_4)_2$ on cracks, as presented in Fig. 7(C). Yang et al. [156] prepared PEO/PA/Ce multilayer coating on AZ31B Mg alloys. Upon corrosive medium penetration into the Mg substrate, the hydroxide ions from corrosion products reacted with Ce^{3+} ions from the broken PEO/PA/Ce coating, forming a new transformation layer on the damaged area (Fig. 7(D)). Besides, a highly amorphous Ce_xO_y coating was designed to seal PEO-coated AZ31B alloy, and the high internal energy drove the diffusion, migration, and precipitation of cerium ions to provide long-term corrosion protection [157].

LDH, also known as hydrotalcite, is an inorganic clay material with metal hydroxide laminates and interlayer anions. The load and release of corrosion inhibitors through the interlayer anion exchange bestows on the coating self-healing functionality. The schematic diagram of the LDH structure

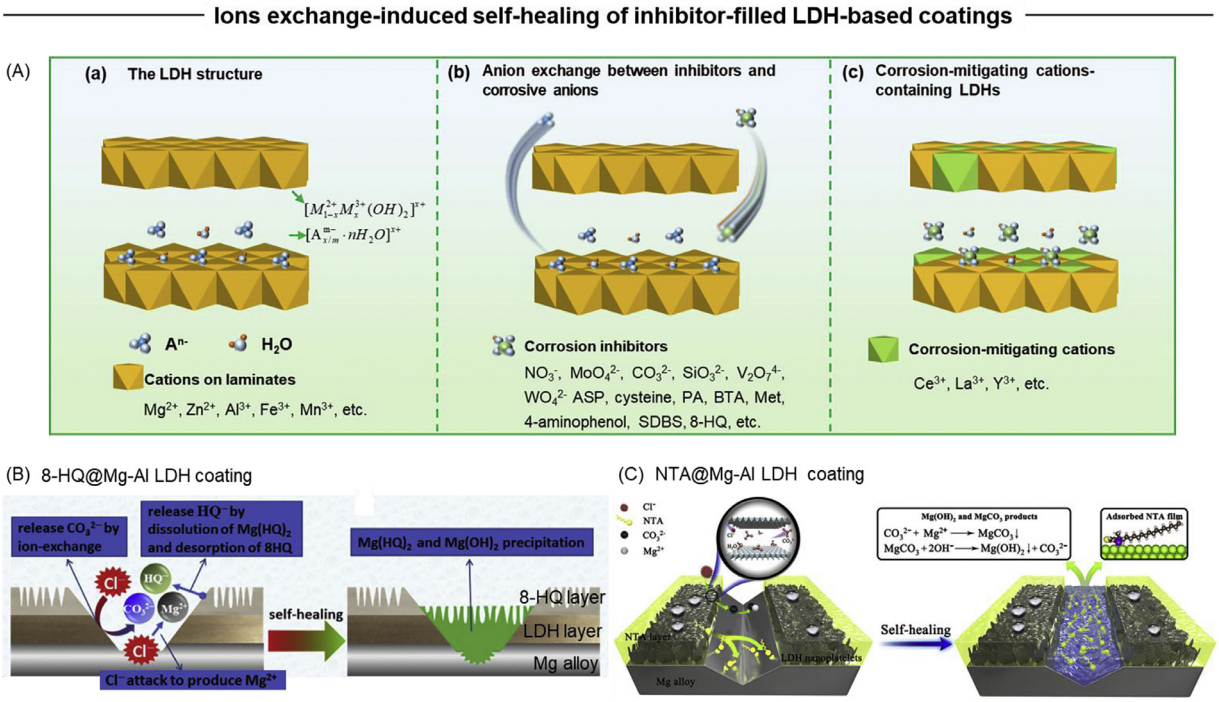


Fig. 8. Schematic diagram of (A) LDH (a) the LDH structure, (b) anion exchange between inhibitors and corrosive anions, and (c) corrosion-mitigating cations-containing LDHs with intercalated inhibitors; (B) the 8-HQ@Mg-Al LDH coating [167]; (C) the NTA@Mg-Al LDH coating [168].

is shown in Fig. 8(A). The general structural formula of LDH is $[M^{II}_{1-x}M_x{}^{III}(OH)_2][A^{n-}]_{x/n} \cdot mH_2O$, where M^{II} and M^{III} represent divalent and trivalent metal cations, respectively; A^{n-} is an interlayer anion, x is the molar ratio denoted with the $M^{III}/(M^{II}+M^{III})$ ratio, normally between 0.17 and 0.33 [158,159]. The composition design of M^{II} and M^{III} endows LDH with the capability of trapping aggressive anions by interlayer anion exchange. When the molar ratio of M^{II}/M^{III} is between 2 and 4, corrosion inhibitors loaded in LDH can exchange with chloride ions [160,161]. As an intelligent container of inhibitors, LDH exhibits dual capacities, including physical barrier protection via its nanosheets, and corrosive anion capture through interlayer anion exchange. The inhibitor-intercalated LDH coating captures aggressive anions via intercalation/deintercalation processes and releases encapsulated inhibitors at the same time [162]. LDH-$WO_4{}^{2-}$ and LDH-$MoO_4{}^{2-}$ were designed to trap the aggressive Cl^- and the substituted $WO_4{}^{2-}$ and $MoO_4{}^{2-}$ reacted with the dissolved Mg^{2+}, forming $MgWO_4/MgMoO_4$ protective film to inhibit the local pitting corrosion [163,164]. Albumin-WO_3 composite was linked with LDH film through hydrogen bonding, exhibiting remarkable corrosion protection through the dopant release, corrosive ion capture, and charge transport inhibition [165].

The inorganic anions prefer to form an insoluble oxide/hydroxide layer, while organic inhibitors tend to generate an adsorption or complex precipitation layer on defect regions, both of which are accompanied by the alteration of double-electrical layer structures on Mg alloy surface. 8-HQ, as an effective organic corrosion inhibitor, is able to protect substrates by forming $Mg(HQ)_2$ precipitates at defects [166]. The combination of 8-HQ and LDH emerges as an attractive candidate for self-healing coatings, significantly improving the anticorrosion performance of LDH-based systems. Apart from the deposition of $Mg(HQ)_2$ precipitation, anion exchange between Cl^- and $CO_3{}^{2-}$ and subsequent transformation to protective $Mg(OH)_2$ also accounted for the great anticorrosion performance of 8-HQ/LDH composite systems [167]. Similarly, the Mg-Al LDH coatings modified with thiophene derivative inhibitors were fabricated on Mg alloys, and physical damages were repaired by forming chemical precipitates and an adsorped film on defects [168]. The self-healing mechanisms of organics (8-HQ, thiophene derivatives (NTA))-filled Mg-Al LDH coatings were demonstrated in Fig. 8(B) and (C), respectively.

Additionally, the liquid coating has been employed for self-repairing the damaged surface, and introducing a slippery liquid-infused porous surface (SLIPS) to protect Mg alloy from corrosion is a promising attempt. A SLIPS usually consists of two parts: a structural porous substrate with free space for loading the lubricant and a low surface tension lubricant [169]. As shown in Fig. 9(A), a SLIPS was constructed in a partition structure on AZ31B Mg alloy, wherein the inherent fluidity feature of the lubricating layer enabled a self-healing capability even under external mechanical damage [170]. Nevertheless, long-term immersion, high temperature, and ultraviolet radiation will lead to the degradation of lubricating oil, thus developing a stable self-healing SLIPS is a promising route. Then, oleogel was infused into the micro/nano structure of the superhydrophobic samarium/myristic acid com-

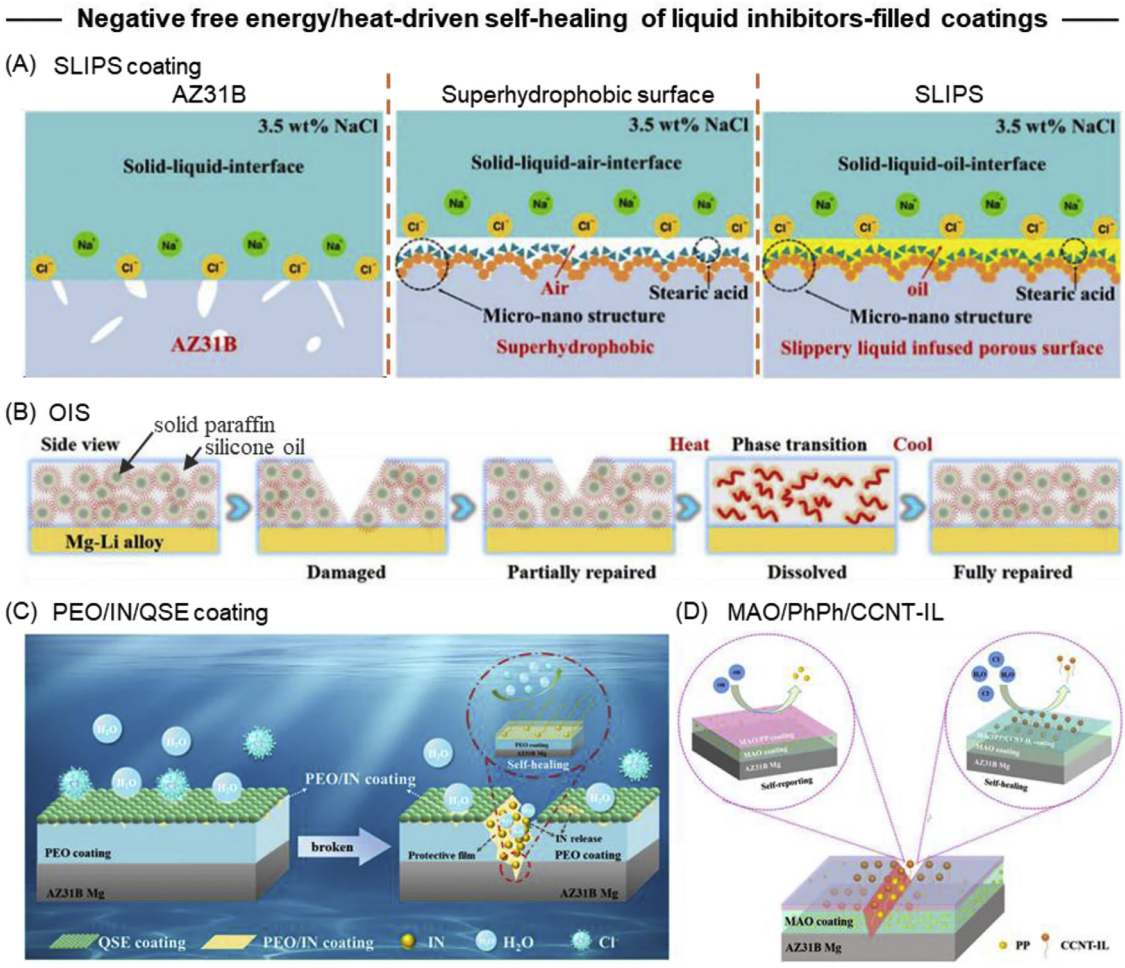


Fig. 9. Schematic diagram of (A) the inherent fluidity-driven self-healing of SLIPS coating [170]; (B) the heat-induced repair process of oleogel infused surface (OIS) [171]; (C) the spontaneous rupture and flow of PEO/IN/QSE coating [175]; (D) the self-reporting (PhPh) and self-healing PEO/PhPh/CCNT-IL coating [176].

plex by a spin-coating method. The regenerated paraffin-based oleogel formed through thermal-assisted phase transformation of paraffin in silicone oil, and then flowed to fill the damaged grooves due to the surface tension effect (Fig. 9(B)) [171].

Ionic liquid (IL) is an organic liquid substance at ambient temperature and is composed of only ions, usually an organic cation and a specific inorganic anion. It was regarded as a class of promising "green" inhibitors due to its distinct properties, including negligible vapor pressure, chemical and thermal stability, and excellent tunable structures [172]. The IL was incorporated into the porous structure of PEO coatings, followed by a sealed topcoat. Once the coating was damaged, IL ruptured driven by thermodynamics and formed an adsorbed film covering damaged regions [173]. Additionally, the hydrophobic feature of long alkyl chains in IL prevented contact and penetration of corrosive ions, enhancing corrosion resistance and service durability [174]. Furthermore, a superhydrophobic topcoat (QSE) was employed to seal the micropores of PEO coating embedded with imidazolium-based IL [175]. The corrosion resistance mechanism involved the barrier effect from the superhydrophobic coating, the adsorbed film on PEO coating formed by imidazolium-based ionic liquid (IN), and the micropore sealing effect via IN (Fig. 9(C)). Similarly, Huangfu et al. [176] incorporated the ionic liquid grafted carboxylated carbon nanotube (CCNT-IL) and phenolphthalein (PhPh) into the PEO coating (Fig. 9(D)). Once the coating was damaged, the released CCNT-IL chelated with Mg^{2+} and then covered the damaged area, thereby repairing the corrosion damage and prolonging the service lifetime.

The incorporation (doping and filling) of corrosion inhibitors into priming coatings on Mg alloys not only grants the coating self-healing functionality, but also infiltrates and seals micropores to improve the coating integrity.

3.2.3. Multi-inhibitor-incorporated self-healing composite coatings

The composite coating combines the excellent adhesion of the base coating and the isolation effect of the top coating, synergistically providing dual-layer protection for Mg alloy. On this basis, the corrosion inhibitors are incorporated into the composite coatings simultaneously, achieving dual self-healing capacities through defect-specific precipitation

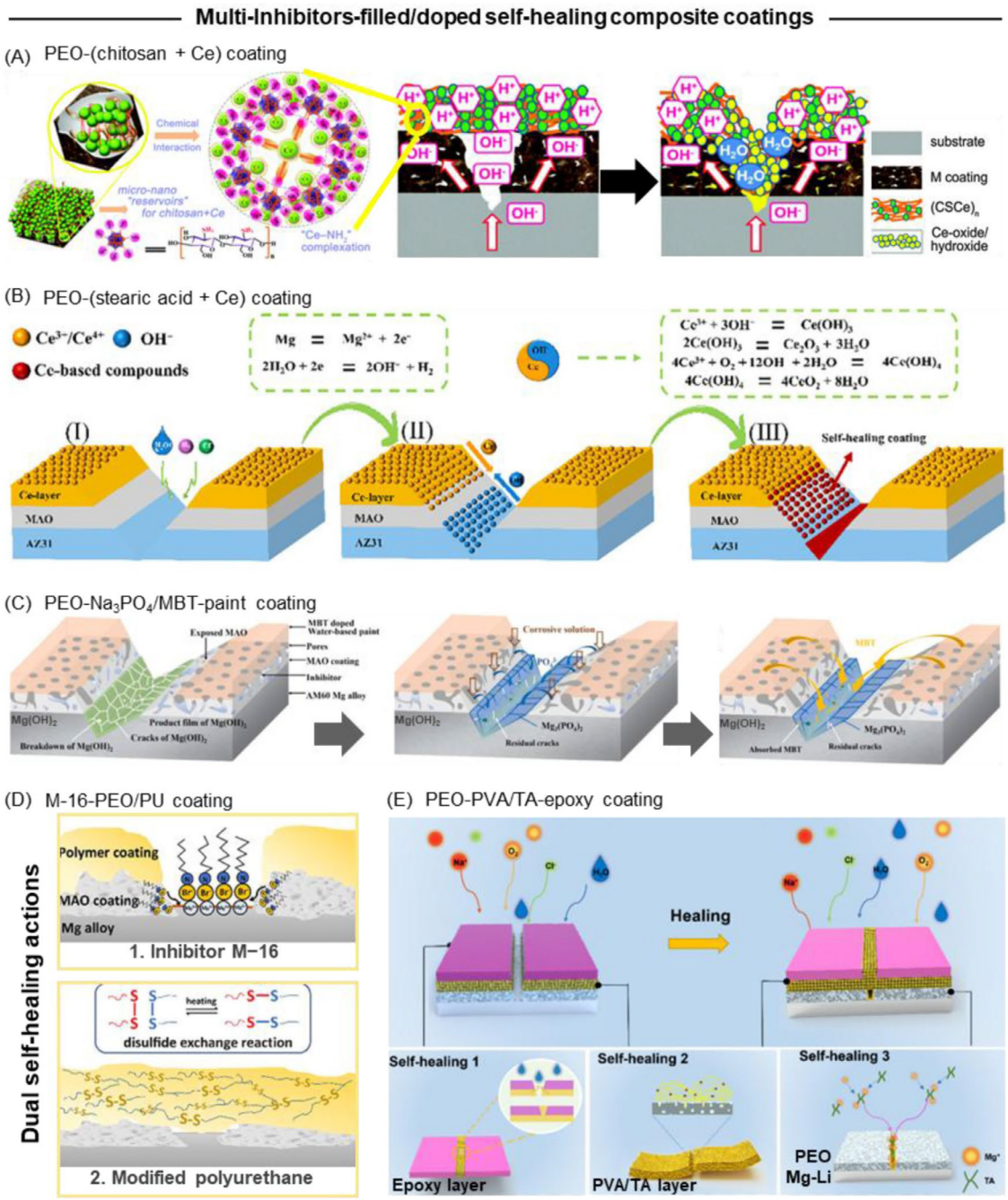


Fig. 10. Schematic illustrations of self-healing mechanisms of multi-inhibitors-doped/filled PEO-based composite coating: (A) the PEO-(chitosan + Ce) coating [178]; (B) the PEO-(stearic acid + Ce) coating [179]; (C) the PEO–Na_3PO_4/MBT-paint coating [184]; (D) the M-16-PEO/PU coating [182]; (E) the PEO-PVA/TA-epoxy coating [183].

and barrier reconstruction. The self-healing mechanisms of inhibitor-filled/doped composite coatings were demonstrated in Fig. 10. In this regard, based on the porous structure of PEO and controlled interlayer intercalation of LDH, a composite smart coating composed of PA-modified LDH layer and Ce-sealed PEO coating was constructed on AZ31 Mg alloy [177]. The PO_4^{3-} anions released from LDH through ion exchange with aggressive Cl^- and the Ce cations released from the PEO coating, which promoted synergistically the formation and coverage of denser insoluble products on the damaged regions. Moreover, the micropores in PEO coatings were utilized as micro-nano reservoirs for organics (chitosan and stearic acid) and Ce cations. For instance, Jia et al. [178] used CeO_2-doped chitosan to seal the PEO-coated Mg alloys, wherein the formation of Ce-NH_2 complexes significantly enhanced interfacial compatibility between the polymer matrix and corrosion inhibitors. The local pH rise triggered the swelling of chitosan and deposition of Ce-containing oxides, thereby inhibiting anodic activation and repairing defects autonomously, as shown in Fig. 10(A). Liu et al. [179]

employed $Ce(NO_3)_3$-doped stearic acid for post-treating PEO coatings, and the Ce ions near the damaged regions were released and migrated to the micro defects, forming a Ce-based protective film to prevent further corrosion (Fig. 10(B)).

In Fig. 10(C), 2-mercaptobenzothiazole (MBT)-sealed Na_3PO_4-filled PEO composite coatings were prepared on Mg alloys, and the released Na_3PO_4 and MBT synergistically enhance the self-healing ability of the coatings via forming protective $Mg_3(PO_4)_2$ and adsorbing on the defects, respectively. Moreover, SMPs and dynamic reversible reconfiguration mechanisms of polymer matrices are integrated into multifunctional self-healing composite coatings. Upon exposure to ultraviolet light and visible light, disulfide bonds undergo radical transfer and cross-reaction [180,181]. A composite coating consisting of a corrosion inhibitor-embedded PEO layer and a disulfide bond-modified polyurethane (PU) topcoat exhibited a synergistic effect on defect sealing: the release of corrosion inhibitor from the PEO coating, the dynamic disulfide exchange reaction, and the thermal-driven shape memory effect of PU [182]. Polyvinyl alcohol (PVA) swelling, H-bonding formation between PVA and tannic acid (TA), and insoluble TA-Mg complex deposit synergistically prevented the expansion of damaged regions on PEO-coated Mg-Li alloys, maintaining coating integrity in harsh, corrosive environments [183].

3.3. Self-healing coatings with a combination of a protective layer, corrosion inhibitors, and containers

Incorporating corrosion inhibitors into self-healing coatings suffers from the uncontrolled release of corrosion inhibitors, leading to limited self-healing cycles and constrained operational durability. The encapsulation of healing agents into micro/nano containers, such as fibers, nanotubes, and capsules, is a promising strategy, allowing the controlled release of active agents to targeted defect areas. The embedded nanocontainers are able to autonomously respond to various external/internal stimuli and release corrosion inhibitors controllably. The common containers, including LDH, GO, mesoporous silica ($MSiO_2$), attapulgite (APT), and halloysite nanotube (HNT), are characterized by large specific surface areas and porous/intercalated structures, which facilitate efficient loading and controlled release.

3.3.1. Corrosion inhibitor containers with ion exchange

LDHs intercalated with corrosion inhibitors are widely employed as fillers for sealing microdefects in PEO coatings. During exposure to corrosive solution, PEO coatings serve as physical barriers, while LDHs simultaneously play triple roles: (1) physical barrier and PEO coating sealants derived from mult-oriented grown LDH nanosheets, and (2) aggressive ions capture via anion exchange with interlayer inhibitors, and (3) the release of intercalated inhibitors and autonomous repair of the damage regions through physical adsorption or complexation reaction-mediated barrier reconstruction [185]. Li et al. [186] intercalated inorganic silicate ions into LDHs on the anodized AZ31B Mg alloys. The released SiO_3^{2-} from the LDH via anion exchange reacted with Mg^{2+} and formed $MgSiO_3$ precipitates in the corroded areas (Fig. 11(A)). Besides, Gnedenkov et al. [187] prepared PEO/LDH/BTA composite coatings on MA8 Mg alloys, wherein the organic ions were intercalated into LDH interlayer. The formation of $Mg(BTA\text{-}H)_2$ complex in the defect zone provided active repair and protection (Fig. 11(B). Besides, inhibitor-incorporated LDHs were also dispersed in epoxy coatings for achieving self-healing function. Due to the stronger adsorption characteristic of methionine (Met) in different pH environments**,** the epoxy coatings containing Met-intercalated Mg-Al LDHs were employed on Mg alloys. The Met ions were released from the Mg-Al LDH interlayers by ion exchange with Cl^- ions, and then they adsorbed on the metal surface by the hydrogen bonding or the direct chemisorption in a neutral/alkaline environment and rapid physical absorption in an acidic environment [188]. In addition, 2D covalent organic frameworks (COFs) with nanofluidic channels were utilized to load corrosion inhibitors. Ag^+ cations and 2-mercaptobenzimidazole (2-MBI) inhibitors were sequentially immobilized on the COF skeleton dispersed in the Polyethersulfone (PES) coating, where 2-MBI inhibitors were linked with Ag^+ ions via coordination bonds. As shown in Fig. 11(C), the porous COF absorbed the aggressive ions and formed silver-containing compounds, preventing the corrosive media from contacting the Mg substrate. Meanwhile, 2-MBI inhibitors linked by silver bridges were rapidly released through nanofluidic channels, forming a passivation film as a corrosion barrier [189].

It is well known that all constituents of LDH encompassing divalent and trivalent cations in laminates and anions in interlayers are inherently controllable. As shown in Fig. 8A(c), corrosion-mitigating cations displaced the native cations on LDH laminates, thereby conferring upon LDH itself the self-healing capacity from inhibiting cation release and deposition. The ternary LDH intercalated with inhibitors shows dual self-healing effects: (1) LDH-derived metallic cation hydroxide precipitation, and (2) the complexation reaction between corrosion inhibitors and Mg^{2+} cations from substrate corrosion or metallic cations from LDH laminates (Fig. 11(D) and (E)) [190,191]. As for organics intercalated ternary LDHs, the released metallic cations (Mg^{2+}, Al^{3+}, and La^{3+}) from LDH laminates could form hydroxide precipitations in a local alkaline environment and complex precipitations with the anion-exchange released 4-aminophenol anions, filling and sealing the corroded regions to prevent Cl^- attacks [192]. Mg-Al-Ce LDH incorporated with sodium dodecylbenzene sulfonate (SDBS) on PEO-coated Mg alloys released Ce^{3+} to form a cerium hydroxide for covering the defects. Meanwhile, the released SDBS inhibitors were electrically adsorbed onto the Mg alloy surface through π-π stacking interactions, blocking further invasion of the corrosive medium [193].

However, initial rapid ion exchange may reduce the crystallinity and the stability of LDHs, hardly providing long-term corrosion protection [194]. GO has strong barrier properties against molecular (O_2 and H_2O) and ion (Cl^-), therefore it

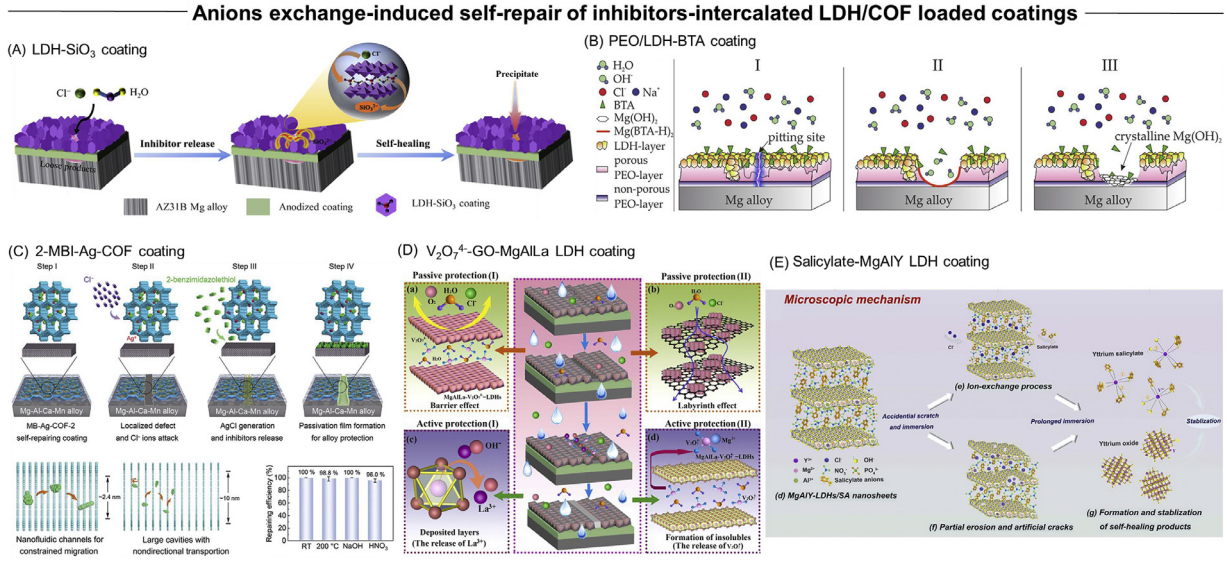


Fig. 11. Schematic illustration of anion exchange-induced self-healing mechanism of (A) the LDH-SiO_3 coating [186]; (B) the PEO/LDH-BTA coating [187]; (C) the 2-MBI-Ag-COF coating [189]; (D) the ternary Mg-Al-La LDH coating doped with vanadate ($V_2O_7^{4-}$)-intercalated GO [190]; (E) the salicylate-intercalated Mg-Al-Y LDHs film [191].

is generally applied to block the penetration of corrosive media into protective coatings and inhibitor containers [195]. Moreover, GO has a greater specific surface area and abundant negative charges, which can adsorb a large number of metallic cations, promoting the in-situ nucleation and providing favorable support for LDH growth [196]. Wu et al. [197,198] incorporated GO and GO fixed inhibitors (8-HQ, SDBS) into LDH interlayers on MAO-coated Mg alloys to ensure the structural stability of the LDHs and the stable existence of corrosion inhibitors for a long time. GO increased the tortuous conduction path of corrosive media, and the released inhibitors complexed with Mg^{2+}, thereby repairing the damaged defects and avoiding the penetration of corrosive media.

Additionally, MXenes, as a class of two-dimensional transition metal carbides/nitrides with high loading capacity, are applied as inhibitor carriers to achieve corrosion-resistant coatings with encapsulated and well-dispersed inhibitors. MXenes have been widely employed in self-healing coatings on Mg alloys, including functionalized MXene doped into coatings, synergistic effects between LDH and MXene [199], and corrosion inhibitor-loaded MXene incorporated into coatings [200]. Apart from the self-healing and protective functions of loaded inhibitors, the barrier effect of MXene and LDH nanosheets, and the LDH-derived metallic hydroxide precipitates in the scratched areas synergistically constructed the self-healing behaviors. Based on the 4-aminophenol intercalated Mg-Al-La LDHs [192], 4-aminophenol-doped MXenes were incorporated into Mg-Al-La LDHs for sealing the PEO coatings [201]. Herein, the newly added MXenes not only optimized the structure of LDH nanosheets to enhance the barrier effect, but also carried more 4-aminophenol inhibitors owing to the polyhydroxy structures of MXenes, achieving the phased release of corrosion inhibitors and La^{3+}.

The cation-exchange containers for corrosion inhibitors are sheet-like and layered phyllosilicates, including zeolites, montmorillonites (MMTs), and their derivatives. It was reported that MMTs were employed as containers for corrosion inhibitors (Ce^{3+}/Zr^{4+}) and dispersed into a hybrid organic-inorganic sol. The released Ce^{3+}/Zr^{4+} cations through ion substitution formed Ce and Zr oxides to cover the defects [202]. Chen et al. [203] incorporated in-situ Ce-loaded zeolite microparticles into PEO coatings on Mg alloys. The cation exchange-induced release of Ce^{3+} from zeolite microparticles leads to the formation of Ce hydroxide/oxide precipitates, enhancing the coating's long-term barrier property against the aggressive solution.

3.3.2. Corrosion inhibitor containers with pH

Generally, metal corrosion processes in aqueous electrolytes are composed of anodic dissolution and cathodic reactions. Cathodic reactions are divided into hydrogen evolution and oxygen reduction reactions, both of which are accompanied by localized pH elevation. Commonly, Mg corrosion is intrinsically coupled with hydrogen evolution reaction, and anodic and cathodic reactions are shown in Eqs. (1–2), respectively. Once hydroxyl ions are produced in the cathodic site, local pH shifting positively to a higher value, pH-sensitive containers undergo structural decomposition and controlled release of corrosion inhibitors for retarding corrosion and healing defects. The frequently used pH-responsive containers for corrosion inhibitors include $MSiO_2$, HNT, APT, and polymer-based microspheres, which leverage environmental pH changes for on-demand release of inhibitors (Fig. 12(A)).

$MSiO_2$ nanoparticles exhibit a high surface area (700–1500 m^2/g), uniform cylindrical pore size (2–50 nm), and large pore volume (about 1 cm^3/g), thereby enabling a sufficiently high uptake and storage of the corrosion inhibitors [204].

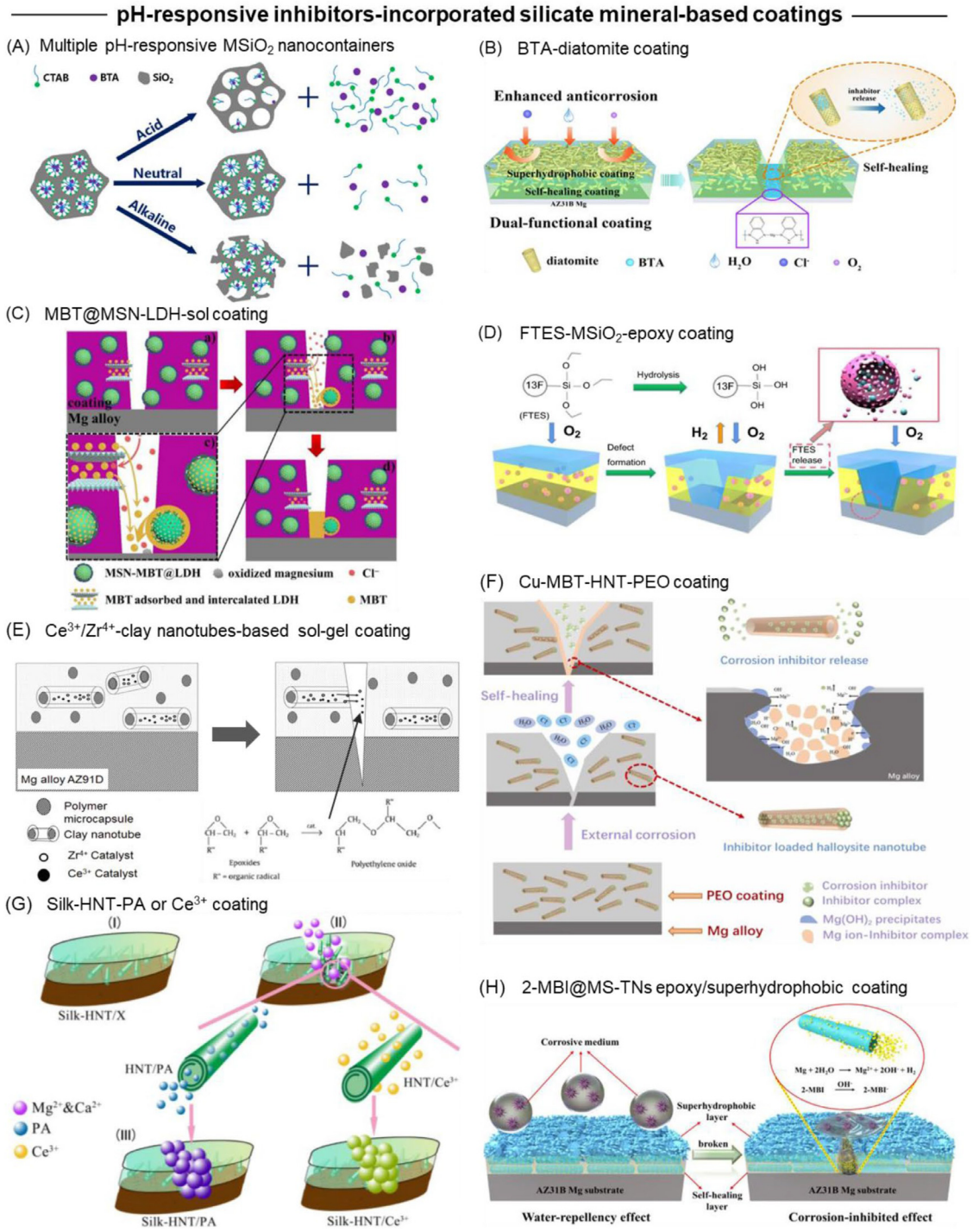


Fig. 12. Schematic representation of self-healing mechanism of pH-responsive silicate minerals loaded with inhibitors: (A) the multiple pH-responsive release of inhibitors from $MSiO_2$ nanocontainers in different pH environments. [205]; (B) the BTA-diatomite composite coating [206]; (C) the sol coating containing MBT@MSN-LDH [207]; (D) an epoxy coating doped with FTES-$MSiO_2$ [208]; (E) the Ce^{3+}/Zr^{4+}-clay nanotubes-based sol-gel coating [210]; (F) the Cu-MBT-HNT-PEO coating [213]; (G) the Silk-HNT-PA/Ce^{3+} coating [215]; (H) the hybrid coating containing a superhydrophobic topcoat and a 2-MBI@MS-TNs epoxy bottom layer [218].

The BTA-loaded diatomite/epoxy resin coating released BTA from the dissolved SiO_2 containers due to the Mg corrosion-derived alkaline environment [205], and then BTA chelated with Mg^{2+} to further hinder the penetration of corrosive ions (Fig. 12(B)) [206]. Furthermore, it was reported that MBT and fluoride ions co-incorporated $MSiO_2$ nanoparticles were doped into Ni coatings on Mg alloys. During Mg corrosion process, the released F^- ions reacted with Mg^{2+} to generate a protective MgF_2 film, while the released MBT simultaneously adsorbed onto the corrosion sites [77,78]. Notably,

the LDH nanosheets were employed as gatekeepers to encapsulate MBT-loaded $MSiO_2$ nanoparticles (MSN), and then the synthesized nanocontainers were incorporated into the sol coatings spun onto AZ31 Mg alloys [207]. As the dissolution of CO_2 and the hydrolysis of Mg and Al cations, the decreased pH value in the anodic zones triggered the disintegration of LDHs, facilitating the release and the subsequent adsorption of MBT molecules onto the defect sites by covalent Mg-S groups (Fig. 12(C)). Besides, the composite 1H, 1H, 2H, 2H-perfluorooctyltriethoxysilane (FTES)-loaded hollow $MSiO_2$ was embedded in an epoxy coating on AZ91D Mg alloy. As the pH value around the anodic area increased, the released FTES from the dissolved $MSiO_2$ subsequently hydrolyzed to produce silanol (Si-OH) groups promptly. The active Si-OH groups covalently interacted with epoxy, Mg alloy, and themselves to deposit a dense hydrophobic silane film onto the corrosion sites (Fig. 12(D)) [208].

HNT with mesoscopic or even macroscopic lumen has been widely used as a host for the loading and controlled release of various guests, including inorganic and organic agents [209]. Adsul et al. [210] encapsulated corrosion inhibitors (Ce^{3+}, Zr^{4+}) into HNTs, and the formation of CeO_2 and ZrO_2 in response to pH variation provided corrosion protection (Fig. 12(E)). Besides, the organic inhibitor BTA-loaded HNTs were also incorporated into PEO coatings on AM50 Mg alloys. The released BTA anions due to pH increase in the corrosion sites adsorbed on Mg alloy surfaces and mediated the precipitation of dense nanocrystalline $Mg(OH)_2$ over the corroded area [211], which was also reported by the other study [212]. To avoid the uncontrolled release of corrosion inhibitors, the insoluble Cu-MBT was designed to block the ports of HNTs loaded with MBT, thereby realizing pH-responsive release of corrosion inhibitors (Fig. 12(F)) [213]. Besides, HNT has a positively charged inner lumen surface and a negatively charged pipe wall, which respectively show electrostatic affinity towards anions and cations [214]. Xia et al. [215] dispersed PA/Ce^{3+} loaded HNTs into a silk coating, exhibiting active and long-lasting protective performance. As corrosion-caused pH increased, the residual random coils and α-helical structure in silk fibroin were stretched due to increased negative charges, consequently releasing HNTs with inhibitors. Subsequently, phytate ions and Ce^{3+} in HNTs were liberated to form Mg^{2+} chelates and Ce-containing precipitates, respectively (Fig. 12(G)).

Attapulgite (APT), as a class of natural inorganic silicate mineral, is employed as a container for loading corrosion inhibitors owing to its rod-like and fibrillar crystalline morphology and micro-nano porous structure [216]. Xue et al. [217] sprayed $Ce(NO_3)_3$-loaded APT on Mg alloys and sealed it with ODA, obtaining a superhydrophobic, long-lasting anti-corrosion composite coating. As the penetration of corrosive species, the cathodic reaction caused the localized pH elevation, which promoted the release of Ce^{3+} from APT and the formation of insoluble cerium oxides and hydroxides for preventing ongoing corrosion processes. Besides, magnesium silicate nanotube (MS-TN) with an uncapped tubular structure has a capacity of loading corrosion inhibitors. Liu et al. [218] fabricated an epoxy resin containing 2-MBI-loaded MS-TNs as the bottom layer on AZ31B Mg alloys, overlaid with a superhydrophobic topcoat. Upon an increased pH due to local corrosion alkalization, the deprotonation and release of 2-MBI in the form of 2-MBI^- thiol or 2-MBI^- thione anion from the uncapped MS-TNs initiated. Subsequently, these species adsorbed onto the metal surface to form a dense protective layer with magnesium oxides for restraining further corrosion (Fig. 12(H)).

3.3.3. Corrosion inhibitor containers with advanced stimuli (pH, light, heat, moisture, corrosion potential, magnetism, mechanical damage)

Apart from the smart containers sensitive to the ionic microenvironment (cations and anions), the novel responsive inhibitor-container systems have been developed, including light, heat, moisture, electrochemistry (corrosion potential), magnetism, and mechanical damage.

Song et al. [219] prepared Ce-based conversion coatings with $La(NO_3)_3$-doped gelatin-chitosan microcapsules on AZ91D Mg alloys. Upon breaking the coating integrity, the microcapsules migrated into the damaged area, triggering the release of La^{3+} to supplement the conversion coating immediately. Except for self-healing capability, smart coatings with self-reporting function have been constructed through the pH-responsive chromogenic reactions involving PhPh and OH^- generated during cathodic reactions. The onset of coating damage, along with the rupture of microcapsules, activates the release of inhibitors from reservoirs to complete the self-healing process. Zeng et al. [220] constructed a smart self-healing coating with an autonomous early corrosion sensing function via embedding PhPh/epoxy resin-loaded urea-formaldehyde (UF) microcapsules into the polyurea resin coating. Defects from the preparation and service process induced the rupture of microcapsules, and the encapsulated PhPh and epoxy resin flowed out along the crack and gradually filled the crack (Fig. 13(A)).

Ding et al. [221] prepared innovatively corrosion potential-responsive sol-gel coatings containing supramolecular assemblies, 8-HQ-loaded magnetic nanovehicles, and disulfide linkers. Notably, the smart nanocontainers gathered nearly the surface of Mg alloys under a magnetic field. Upon the Mg corrosion potential being exerted, the cleavage of disulfide linkers led to the separation of supramolecular assemblies from the nanocontainers and the instant release of 8-HQ, facilitating the quick formation of the compact protective molecular film (Fig. 13(B)). In addition, Zhang et al. [222] synthesized conductive poly(3,4-ethylenedioxythiophene) (PEDOT) nanocapsules loaded with 8-HQ inhibitors, and the declined open circuit potential due to Mg corrosion stimulated the redox-responsive of the PEDOT nanocapsules. Accompanied by the breakage of hydrogen bonds and covalent bonds between 8-HQ and PEDOT, 8-HQ inhibitors were released and formed $Mg(HQ)_2$ passive layer (Fig. 13(C)).

A composite PU coating doped by Met-functionalized MXene was fabricated on Mg alloy (f-MXene@PU). The coating possessed triple effects of active (inhibiting effect) and passive

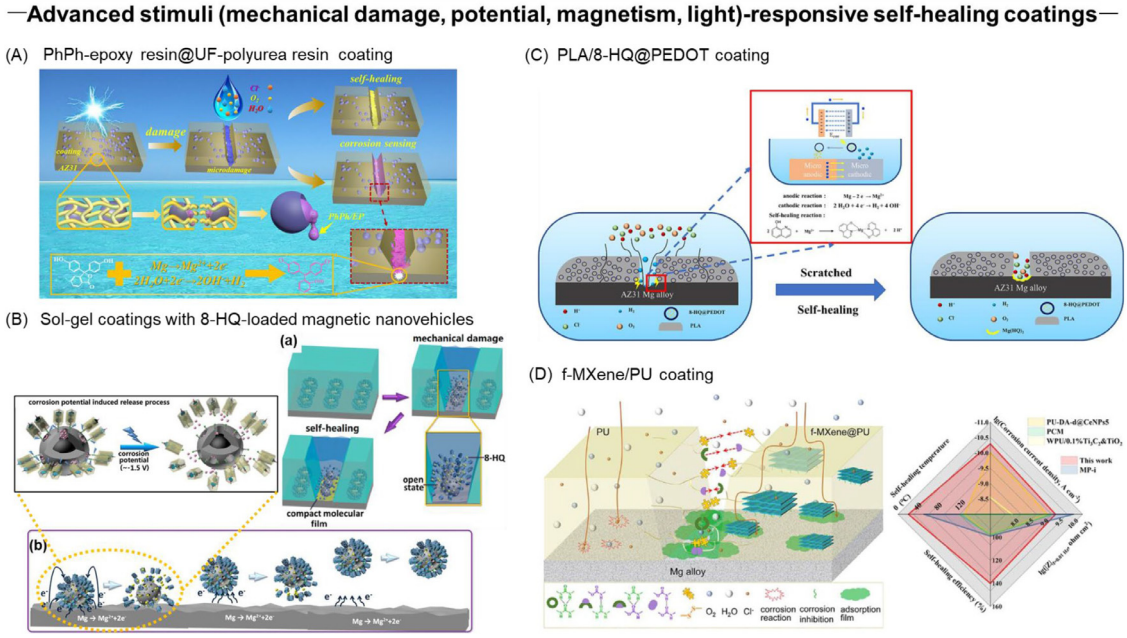


Fig. 13. (A) The corrosion sensing (PhPh) and spontaneously flowing inhibitor-induced self-repair in polyurea resin coating containing PhPh/epoxy resin-loaded UF microcapsules [220]; (B) corrosion-potential-responsive self-healing mechanism of the functional sol-gel coating: (a) with the aid of magnetic field, (b) the release process of 8-HQ from inhibitor systems close to/far away from the substrate surface [221]; (C) corrosion potential-responsive self-healing mechanism of the PLA/8-HQ@PEDOT coating [222]; (D) the visible light-induced self-repair of f-MXene/PU coating [223].

(barrier performance) anticorrosion and self-healing functions (Met adsorption film). As exhibited in Fig. 13(D), the Met adsorption on the anode site and the visible light-induced radical transfer and cross-reaction of PU dynamic disulfide bonds jointly conferred the coating with a superior long-term lifetime [223].

3.3.4. Corrosion inhibitor containers with multiple stimuli

As previously mentioned, the controlled release of corrosion inhibitors is paramount for highly efficient self-healing behavior and multifunctional integration, thus more and more multi-responsive self-healing inhibitor-coating systems have been put forward.

A hybrid coating with water-enabled self-healing and pH/NIR-responsive drug delivery capacity was constructed on WE43 Mg alloy for the clinical application of Mg-based biomedical implants. Polyelectrolyte multilayered coatings were capable of self-healing in water due to dynamic reversible reactions of multilayered hydrated polymers [224]. On the one hand, the high mobility of PAA and branched PEI polyelectrolytes, dynamic reversible hydrogen bonding, and amide bond interactions fostered the water-enabled self-healing behavior. On the other hand, the drug release of PDA-modified drug-loaded microspheres was controlled by pH and photothermal stimuli for prolonging the release of drugs [225]. Additionally, the research team also developed a hyaluronic acid/PEI layer-by-layer assembly coating containing paeonol-encapsulated $MSiO_2$ nanoparticles coated with PDA (PMP). In an acidic environment, the catechol groups of PDA, inhibitor molecules, and silica particles possessed the same charge, resulting in electrostatic repulsion within the nanocontainer and the peeling of PDA coating partially from the $MSiO_2$ surface. Consequently, paeonol inhibitors were released and formed paeonol-Mg complexes to seal the defects. Meanwhile, the degraded coatings and generated defects initiated the diffusion and rearrangement of polymer chains, followed by the dynamic reversible hydrogen bond reformation between hyaluronic acid and PEI (Fig. 14(A)) [226].

Li et al. [227] designed a bio-inspired superhydrophobic three-layer structure coating consisting of the uppermost La-LDH powder film, the intermediate PDMS, and the underlying La-LDH film. The dual self-healing mechanisms were ascribed to the Cl^-/laurate ion exchange process on demand and the migration and recombination of PDMS C–H chains through thermal motion for the minimization of surface free energy (Fig. 14(B)). Besides, a micro/nano-bilayer phosphate-loaded composite coating, comprising a mid-layer LDH with the anion-exchange ability and a top-layer pH-responsive $MgHPO_4$, was constructed on PEO-coated Mg alloys. It exhibited dual stimuli-responsive release of phosphate via alkaline pH activation and ion exchange for self-healing [228]. Ding et al. [229] reported that 2–hydroxy-4–methoxy-acetophenones (HMAPs) were encapsulated into mechanized silica nanoparticles (MSNPs) containing $MSiO_2$ nanoparticles and water-soluble pillar [5] arene-based supramolecular nanovalves, and the MSNPs system released HMAPs due to Mg^{2+} or OH^- stimuli to remedy the corrosive micro-region. Grewal et al. [230] constructed a triple trigger-responsive PA encapsulated PCL hybrid coating and confirmed that water ingress, pH rise, and Mg ion enrichment all triggered PA release from PCL microcapsules, subsequently forming dense HA mineralization in structural defects (Fig. 14(C)).

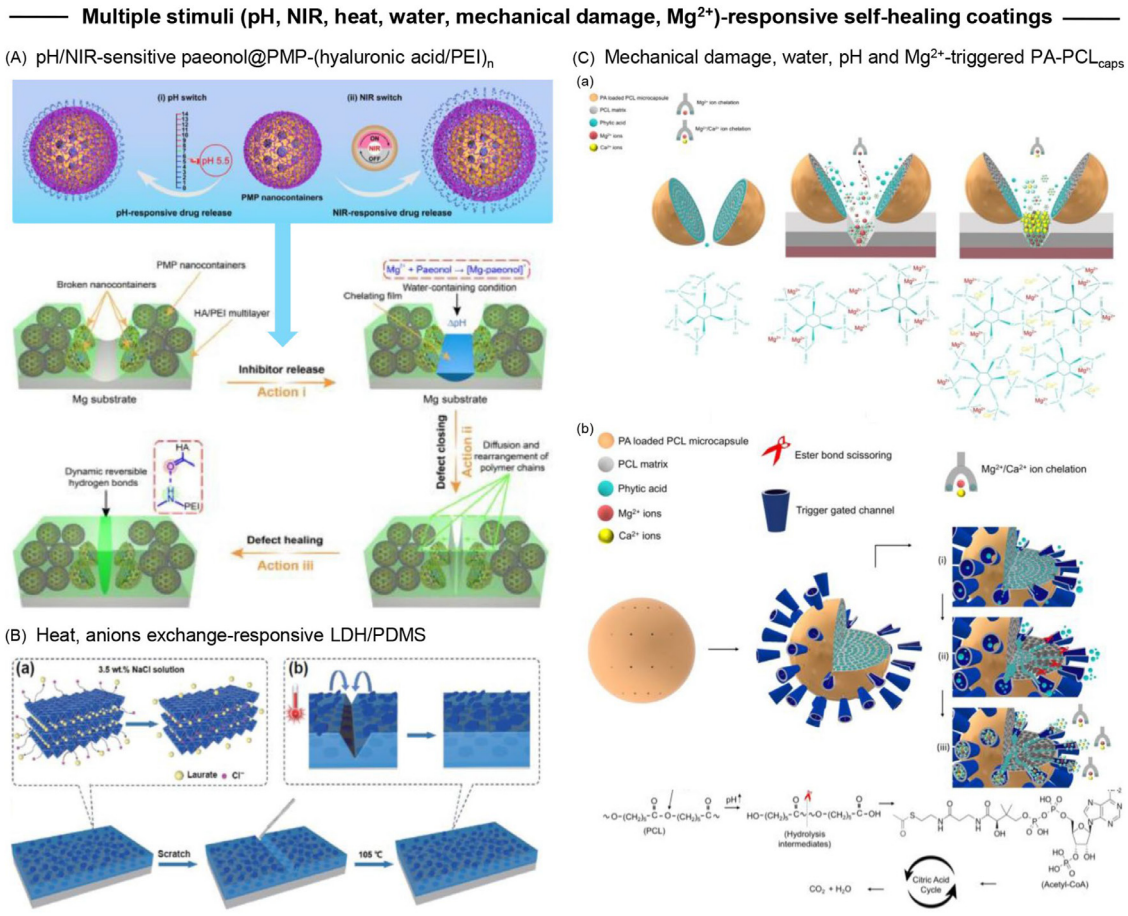


Fig. 14. (A) Schematic illustration of the pH/NIR-responsive self-healing mechanisms of the paeonol@PMP-(hyaluronic acid/PEI)$_n$ coating [226]; (B) dual anion-exchange and heat-activated self-healing mechanisms of superhydrophobic Mg-Al-La LDH/PDMS coating: (a) anion-exchange process between corrosive Cl^- and corrosion-mitigating La ions, (b) the migration of inhibitors to damaged area and recombination of heat-activated C–H chains [227]; (C) stimuli-responsive release of PA from PCL microcapsules of PA-PCL$_{caps}$ coating: (a) mechanical damage, (b) moisture, pH and Mg^{2+} [230].

Multi-stimuli responsive inhibitor-container systems, integrating light, heat, pH, and moisture-sensitive triggers, enable precise control over corrosion protection and autonomous defect repair through the synergistic effects between encapsulated corrosion inhibitors and dynamically formed protective layers.

4. Evaluation methods of self-healing functions

The investigation of precise detection and evaluation methodologies for self-healing behavior and coating performance is critical for developing and applying self-healing systems. The evaluation of self-healing coatings for corrosion protection includes defect construction, microstructure evaluation, and corrosion resistance monitoring, as illustrated in Fig. 15.

Introducing an artificial defect and characterizing the active responsiveness of the coating constitute fundamental procedures of self-healing function evaluation. A standard practice involves introducing mechanical damage to coated samples, followed by immersion in a simulated electrolyte. Artificial defects can be introduced through various ways, including single or cross microstructure damages from scalpels, razors, and nanoindenter systems [146], or chemical damage from O_2-plasma etching [127]. Recently, stress-assisted coating failure was introduced as a damage action to confirm the possibility of self-healing coating to retard SCC [74]. Additionally, the immersion time depends upon the solution medium and the utilized corrosion inhibitor composition [231,232]. The evaluation of the self-healing function requires preliminary manual prefabrication of scratches on coatings. However, a standardized criterion for the dimensions (width, depth, and length) of scratches has not been established to judge whether the coating systems possess self-healing capabilities. Besides, the time length criteria of self-healing response and termination have not yet been determined.

The morphology, composition, and phase of the damaged area after immersion are commonly characterized utilizing nonelectrochemical techniques, including optical microscopy (OM) [220], scanning electron microscopy (SEM) [233], energy dispersive spectroscopy (EDS) [234], atomic force microscopy (AFM) [235], X-ray diffraction (XRD) [236], Fourier transform infrared spectroscopy (FTIR) [136], and micro-computed tomography (micro-CT) [237]. Nonelec-

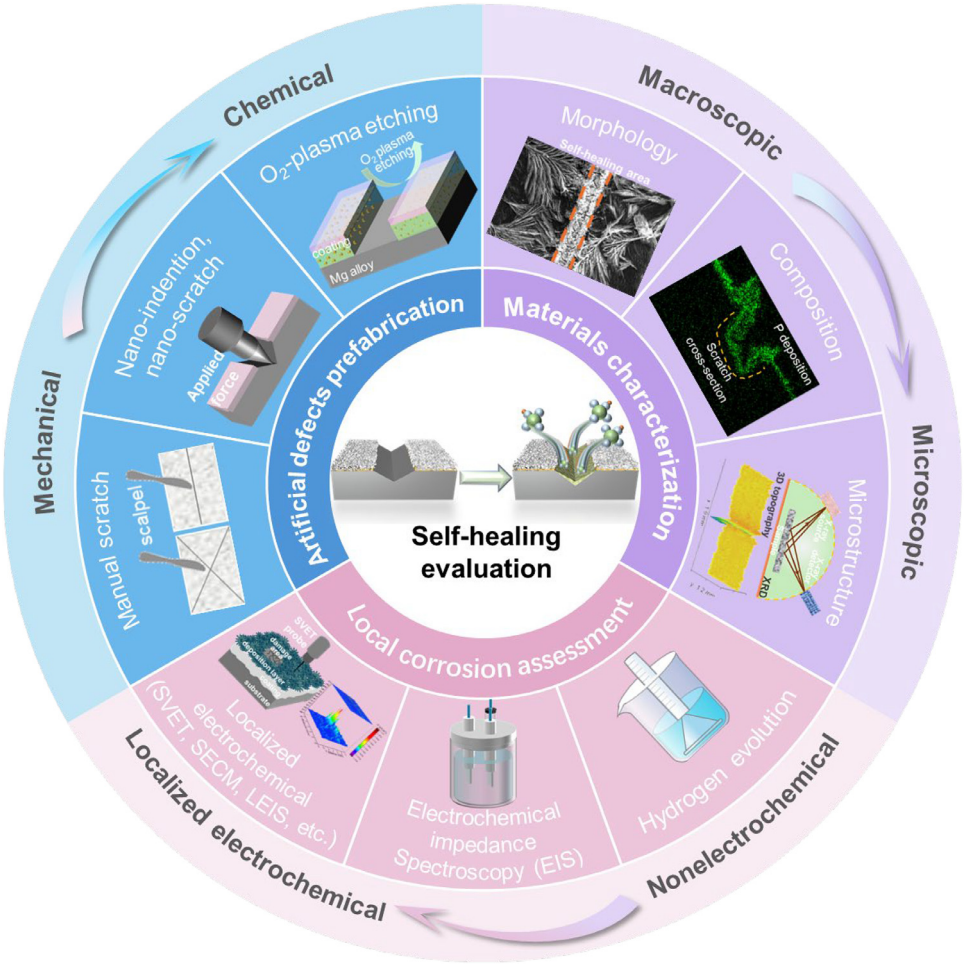


Fig. 15. Evaluation methodologies of self-healing mechanisms and protective efficacy of coatings on Mg alloys.

trochemical techniques for corrosion evolution, such as hydrogen evolution (HE), salt spray testing, weight loss (WL) measurements, inductively coupled plasma mass spectrometry (ICP-MS) [149], are usually utilized to verify synergistically the self-healing functions of coatings. The scratch length and width decreased before and after healing, and the deposit covering the scratch contains the corrosion inhibitors, indicating that self-healing behavior has occurred. The slowing of cathode corrosion rate via HE, the release of inhibitors using ICP-MS, and the weight gain of the sample through WL synergistically verify the occurrence of self-healing behavior. OM provides real-time visualization of healing products covered on the scratched region, whereas more precise SEM and EDS enable further investigation about morphology characterization and composition identification of coatings during sequential healing stages. Moreover, AFM with exceptional resolution can be utilized to characterize 2D and 3D topography of the cracks in liquid-infused gels. XRD and FTIR can elucidate the structural evolution of self-healing systems through monitoring crystallinity alterations and functional group transformations by shifts in peak positions or by changes in peak intensities, such as the dissolution and recrystallization of LDH, the rupture and recombination of diverse reversible bonds of polymers during the self-healing process.

Generally, nonelectrochemical and electrochemical techniques support abundant evaluation information about the self-healing capability of coatings from macro and micro perspectives. Electrochemical techniques are utilized to study surface activity and autonomous repair behaviors of coatings, which are combined with morphology characterization and phase identification to infer and verify the self-healing mechanism. Among the electrochemical techniques, electrochemical impedance spectroscopy (EIS) provides instrumental support for understanding the self-healing mechanism from a macro perspective [166,223]. The self-healing efficiency of the scratched coating relative to the unscratched coating can be calculated using Eq. (5) [113,223]:

$$\eta = \frac{Z^{i}_{f_{low}}}{Z^{0}_{f_{low}}} \times 100\% \tag{5}$$

in which $Z^{0}_{f_{low}}$ represents the low-frequency impedance value of the unscratched coating, and $Z^{i}_{f_{low}}$ represents the low-frequency impedance value after the scratched coating has been repaired for i h.

Furthermore, localized electrochemical techniques such as scanning vibrating electrode technique (SVET) [149,183], scanning electrochemical microscopy (SECM) [182], local-

ized electrochemical impedance spectroscopy (LEIS) [238], scanning Kelvin probe (SKP) [136] and scanning ion-selective electrode technique (SIET) provide direct and in-depth insights into self-healing behavior by detecting dynamic repair process at local defects. The topography and local reactivity of a local crack-containing region of the coating reveal the amount of corrosion with different colors from blue (high corrosion protection) to red (a damaged area by corrosive elements). If localized electrochemical characterization after healing shows a reduction of the corrosion rate in the crack, the self-healing behavior is successfully proved. To obtain comprehensive and in-depth investigations of localized corrosion and repair processes, multiple scanning probe techniques were employed together [144]. Moreover, a combination of slow strain rate stress corrosion creep and electrochemical potential test was designed to evaluate dynamic self-healing behavior by detecting coating potentials which reflected the competition results of the crack propagation and repair processes [239].

In particular, scanning probe techniques are frequently employed to monitor the evolution of the electrochemical activity of artificial defects on coatings to illustrate the active corrosion protection mechanism of coatings. However, the assessment of the electrochemical activity of composite microstructures was constrained by SVET, LEIS, and SKP due to insufficient chemical resolution. Although SECM demonstrates significant advantages in such analyses, the probe measurement is difficult to approach the surface owing to the adverse effect of H_2 evolution. Moreover, it is critical to further improve the spatial resolution of these techniques for characterizing the electrochemical activity of refined microstructures.

5. Overlook

This review generally summarizes Mg corrosion and protective coatings and focuses on discussing active self-healing coatings on Mg alloys. The self-healing coatings are classified into self-healing protective coatings, inhibitor-loaded self-healing coatings, and self-healing coatings incorporated with inhibitor-loaded micro/nano containers. Moreover, remarkable advances have also emerged in stimuli-responsive mechanisms, progressing from single-stimulus response to synergistic multiple-response activation. Based on the above, there are many promising possibilities for self-healing coatings on Mg alloys in the future, as follows.

(1) The self-healing coating on Mg alloy is a new type of surface coating integrating protection and function, which has received extensive attention. At present, the imperfect theoretical system and inconsistent evaluation standards of self-healing coatings are the key scientific issues to be solved urgently in the design and research of self-healing coatings. Furthermore, the core constituents of coatings (inhibitors and containers) require further systematic optimization to develop novel and efficient active materials while achieving high-loading capacity and responsive efficacy.

(2) The self-healing coating systems design tends to be complicated and diversified, and multi-layer and multi-function coatings are becoming research hotspots. However, from the perspective of coating application, the coating process should evolve towards simplification and standardization. Moreover, the interlayered interfacial compatibility demands careful consideration in coating system design. Hence, the advancement of novel self-healing coatings must address not only functional design considerations but also the processing methodology of the coatings.

(3) Extending to the design and research of medical self-healing coatings, it is imperative to further consider the biosafety and biocompatibility of coatings. Significant advancements have been achieved in developing self-healing coatings for medical Mg alloys. Nevertheless, throughout the research process of coatings, medical self-healing coatings primarily focused on material research, with limited in-depth reports on the biological efficacy and self-healing functional properties in vitro and in vivo. More in vitro and in vivo biological experiments need to be carried out for further evaluation of the clinical service characteristics of medical self-healing coatings.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xingxing Zhou: Writing – review & editing, Writing – original draft, Investigation. **Jie Xin:** Writing – original draft, Investigation. **Cheng Wang:** Investigation, Conceptualization. **Kun Qian:** Investigation, Formal analysis. **Xuewei Tao:** Investigation, Formal analysis. **Zhixin Ba:** Supervision, Project administration. **Feng Xue:** Supervision, Investigation, Conceptualization. **Jing Bai:** Supervision, Investigation, Conceptualization. **Bertram Mallia:** Investigation, Formal analysis. **Qiangsheng Dong:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Conceptualization.

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